Can Large Language Models Estimate Willingness to Pay? Evidence from Two Contingent Valuation Replications

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Abstract

We assess whether large language models (LLMs) can substitute for human respondents in referendum-style contingent valuation (CV). Using individual-level data from two published CV studies—a U.S. National Clean Energy Standard (Aldy, Kotchen, and Leiserowitz 2012) and a North Carolina hemlock woolly adelgid control program (Giguere, Moore, and Whitehead 2020)—we construct synthetic personas one-to-one from each respondent's demographic and belief responses, prompt ten frontier and mid-tier LLMs to act as those respondents under the original studies' randomization and bid assignments, and compare synthetic willingness to pay (WTP) to the published benchmarks of \$162 and \$56.53 per household per year. Naive synthetic CV does not recover these benchmarks: most models produce WTP estimates that are negative, near zero, or off by an order of magnitude. The deviations are structured rather

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than random. Synthetic approval-bid curves are systematically flatter than empirical curves across both studies and almost every model, with mechanical consequences for logit-implied WTP. Across-LLM heterogeneity is large and is not rescued by ensemble averaging, and the marginal value of belief inputs is highly context-dependent across the two studies. We argue that synthetic CV is best understood as a form of meta-analytic benefit transfer rather than a substitute for original CV, and identify three near-term applications in regulatory analysis: threshold analysis, sensitivity testing of conventional benefit transfer, and pilot design.

1 Introduction

Contingent valuation (CV) remains the workhorse method for estimating willingness to pay (WTP) for non-market environmental goods, and three decades of methodological development have established the practices that distinguish defensible CV from the methodologically lax studies that the NOAA panel warned against (Arrow et al. 1993; Johnston et al. 2017). A well-designed CV study is, however, slow and expensive. A nationally representative referendum-style survey adequate for benefit-cost analysis typically requires $[**XXX,XXX**] in funding and [**6**–**12**] months of design, sampling, and analysis time (Johnston et al. 2017; Champ, Boyle, and Brown 2017). Federal agencies developing regulatory impact analyses on regulatory timelines rarely have either resource to spare, with the consequence that non-market benefits are routinely set aside as "non-quantified factors" rather than incorporated into headline net-benefit calculations.

A growing literature in marketing research (Brand, Israeli, and Ngwe 2023; Li et al. 2024) and political and social science (Argyle et al. 2023; Bisbee et al. 2024; Horton, Filippas, and Manning 2023) proposes that large language models (LLMs) might serve as "synthetic respondents" for survey instruments—generating responses to prompts in seconds that approximate the responses a human would give to the same prompts. If that approximation is reasonable, applying the idea to CV would let analysts produce WTP estimates in hours rather than months, at a small fraction of the cost of an original study. The substantive question is whether the approximation is in fact reasonable for stated-preference elicitation of non-market values—a question that, despite a small but growing set of related applications, has not been systematically addressed on published CV instruments with known human-survey benchmarks.

We address it through synthetic replications of two published CV studies. Aldy, Kotchen, and Leiserowitz (2012) estimate WTP for a U.S. National Clean Energy Standard (NCES) requiring 80% clean electricity by 2035, using a nationally representative referendum-style survey of 1,010 respondents. Giguere, Moore, and Whitehead (2020) estimate marginal WTP for hemlock woolly adelgid (HWA) control in North Carolina public forests, using an online panel of 907 respondents under both bounded and random-cost choice formats.[1] The studies span substantively different policy domains, sample frames, and elicitation formats, providing a useful cross-context test of synthetic CV. For each study we reconstruct synthetic personas one-to-one from the original respondents' demographic and belief responses, prompt ten frontier and mid-tier LLMs to act as those respondents under the original studies' exact randomization and bid assignments, and analyze the resulting synthetic responses using the original studies' econometric specifications. We then compare synthetic WTP estimates to the published benchmarks of $162 per household per year (Aldy et al.) and $56.53 per household per year (Giguere et al.).

We make three contributions. The primary contribution is empirical. We provide what is, to our knowledge, the most systematic assessment to date of whether LLM-generated responses can substitute for human responses in referendum-style CV: two published studies with known human-survey benchmarks, ten LLMs spanning the current frontier, three persona information conditions, and a one-belief-at-a-time decomposition that isolates the marginal contribution of each non-demographic survey item. The findings, summarized in the next four paragraphs, establish a set of empirical regularities that any future application of LLMs to stated-preference welfare analysis will need to contend with. We organize the assessment around four such regularities.

First, naive synthetic CV does not deliver point estimates of WTP that match human-survey

[1] We replicate Giguere, Moore, and Whitehead (2020) in part because it appears in this journal and is likely familiar to many readers; the headline marginal WTP we benchmark against is reported there as $56.53 per household per year.

benchmarks. In Study 1, [**seven of the ten**] models we evaluate produce negative or near-zero WTP estimates in the demographics-only condition; in Study 2, raw synthetic WTP varies by an order of magnitude across models and choice formats. This holds for frontier models and for mid-tier models, for models with explicit reasoning configurations and for those without, and across the two studies' distinct policy domains. An equally weighted ensemble across all ten LLMs does not rescue the estimate, because the dominant source of error is correlated across models.

Second, the deviation from human-survey benchmarks has a structured character. Across both studies and almost all models, synthetic approval-bid curves are visibly flatter than the empirical curves: synthetic respondents are less price-sensitive than human respondents at every bid level. Because logit-implied WTP is the ratio of the constant to the negated bid coefficient, this slope flatness mechanically inflates or distorts WTP whenever the level of approval is reasonably tracked. We document this pattern descriptively—an observed regularity with a known mechanical implication—and leave the diagnosis of *why* synthetic bid-response curves are flat to subsequent work.

Third, across-LLM heterogeneity is large and is not eliminated by aggregation. The gap between the best- and worst-performing models on any given metric is consistently larger than the gap between demographics-only and demographics-plus-beliefs conditions for a fixed model. Aggregation reduces variance but not bias, and the bias from slope flatness appears to be correlated across models because the underlying training distributions overlap heavily. Practitioners should treat LLM selection as a first-order design decision rather than as a robustness check.

Fourth, the marginal value of belief inputs is highly context-dependent. In Study 1, demographics-only personas produce mostly negative WTP estimates and belief inputs are essentially required to recover plausible levels. In Study 2, the pattern partially reverses: demographics-only personas perform reasonably well in the random-cost design, and adding beliefs sometimes worsens fit. Within Study 1, a single well-chosen belief question often outperforms the full belief battery, suggesting that some belief items carry signal while others actively introduce noise. The lesson for synthetic CV practice is that the relationship between belief richness and synthetic accuracy is not monotonic and is not predictable a priori from study design alone.

These regularities have implications for how the broader LLM-as-respondent literature has framed evaluation. Existing applications (Brand, Israeli, and Ngwe 2023; Horton, Filippas, and Manning 2023; Aher, Arriaga, and Kalai 2023) have largely focused on whether LLMs can recover human-survey responses on average, treating deviations as residual noise. The structured nature of the deviations we observe— particularly the slope flatness—suggests that average-response benchmarks are inadequate for evaluating LLMs as substitutes for human respondents in welfare analysis, where the slope of a response curve enters the welfare measure directly. Diagnosing the slope-flatness pattern at the model layer, and developing corrections that operate on it, are natural priorities for follow-up work.

Our second contribution is conceptual. We argue that the most natural intellectual home for synthetic CV is the benefit transfer literature, not the primary CV literature. Benefit transfer asks how WTP estimates from existing studies should be extrapolated to target contexts where no original study has been conducted, and existing methods include unit value transfer, function transfer, and meta-analytic transfer (Rosenberger and Loomis 2017; Loomis 2015; Boyle et al. 2010). Synthetic CV is, viewed this way, a form of meta-analytic benefit transfer in which the source corpus is the universe of text on which the LLM was trained, and the slope flatness we document parallels known attenuation biases in conventional meta-analytic transfer (Bergstrom and Taylor 2006). This reframing places more reasonable expectations on synthetic CV: it should be evaluated against benefit-transfer benchmarks (transfer errors of [**20–40%**] are typical) rather than against the standards of an original study, and the validity criteria of the benefit transfer literature—generalizability tests, convergent validity checks, transfer error metrics—should be brought to bear on synthetic studies.

Our third contribution is practical. Under the benefit-transfer frame, our results suggest that current synthetic CV is not yet competitive with conventional benefit transfer for direct value substitution, but is plausibly already useful for three more circumscribed purposes: threshold analysis (establishing whether stakeholders' WTP for a non-market good plausibly exceeds a regulatory break-even value), sensitivity testing of conventional benefit transfer (assessing whether a transferred value is plausible given local population characteristics), and pilot design for planned conventional CV studies (identifying wording, ordering, and treatment-arm problems before fielding). We do not find that current synthetic CV is suitable for direct substitution in net-benefit calculations where regulatory decisions hinge on precise WTP point estimates. We develop these recommendations in Section 6.

The remainder of the paper is organized as follows. Section 2 reviews the technical material a reader needs to follow our methods: the structure of a referendum-style CV experiment, the logit specification for WTP recovery, and the basics of how an LLM's persona-conditioned response is generated. Section 3 describes the two replicated studies. Section 4 describes synthetic persona construction, prompt design, model selection, simulation infrastructure, and econometric estimation. Section 5 reports the four regularities summarized above. Section 6 develops the benefit-transfer reframing and the implications for regulatory practice, and Section 7 concludes.

2 Background

This section reviews the technical material a reader needs to follow Sections 4 and 5. Readers familiar with referendum-style CV and logit-based WTP recovery may skim Sections 2.1 and 2.2 and read Section 2.3 in full.

2.1 Referendum-Style Contingent Valuation

A referendum-style CV question presents the respondent with a hypothetical policy and a randomly assigned cost (the "bid"), and asks whether the respondent would vote in favor of the policy at that cost. The format is the modal stated-preference instrument in modern CV practice for two reasons. It is incentive compatible under plausible assumptions about how respondents interpret the elicitation (Carson and Groves 2007), and it produces binary response data that map directly onto the discrete-choice econometric framework that has been the workhorse of stated preference analysis since Hanemann (1984). The two studies we replicate use referendum-style elicitation: Aldy, Kotchen, and Leiserowitz (2012) use a single-bounded referendum with eight bid levels, and Giguere, Moore, and Whitehead (2020) use a choice-experiment generalization in which respondents face referendum-style support questions over policies with multiple randomized attributes including bid.

The structural role of bid in this design is worth emphasizing. Because respondents at higher bids should be more likely to vote no, the slope of the share-approving curve in bid space identifies the price coefficient that anchors the WTP calculation. Most other features of the response distribution—the level of approval, the sensitivity to non-bid attributes, and the variance across respondents—are essentially nuisance parameters from the welfare-recovery perspective.

2.2 Logit Specification and WTP Recovery

The standard analytic framework treats respondent i as comparing the utility of the policy (U_i^P) to the status quo (U_i^{SQ}) and voting in favor when the difference is positive:

$$U_i^P - U_i^{SQ} \;=\; \alpha + \mathbf{x}_i'\boldsymbol{\gamma} - \beta_b \cdot b_i + \varepsilon_i, \tag{1}$$

where $\mathbf{x}_i$ is a vector of respondent characteristics, $\beta_b > 0$ is the marginal disutility of the bid, and ε_i is a logistically distributed error term. The probability of a yes vote is then

$$\Pr(y_i = 1 \mid \mathbf{x}_i, b_i) \;=\; \frac{\exp(\alpha + \mathbf{x}_i'\boldsymbol{\gamma} - \beta_b \cdot b_i)}{1 + \exp(\alpha + \mathbf{x}_i'\boldsymbol{\gamma} - \beta_b \cdot b_i)}, \tag{2}$$

and the parameters $(\alpha, \boldsymbol{\gamma}, \beta_b)$ are recovered by maximum likelihood. Mean WTP across the sample is then

$$\widehat{\mathrm{WTP}} \;=\; -\frac{\hat{\alpha} + \overline{\mathbf{x}}'\hat{\boldsymbol{\gamma}}}{\hat{\beta}_b}. \tag{3}$$

The structural point worth carrying forward is that mean WTP is a ratio in which the bid coefficient appears in the denominator, and proportional errors in $\hat{\beta}_b$ produce inversely proportional errors in $\widehat{\mathrm{WTP}}$. If a synthetic respondent population produces a bid-response slope half the empirical slope, mean WTP is inflated by a factor of two even when every other feature of the response distribution is recovered correctly. This sensitivity is why we evaluate synthetic responses along both level and slope dimensions in Section 5: a synthetic instrument can match human-survey approval shares to within a few percentage points and still produce WTP estimates that are off by a factor of two or more.

2.3 LLMs as Persona-Conditioned Response Generators

A large language model is a probability distribution over sequences of tokens, parameterized by a neural network and trained to predict the next token given a preceding context. [2] For our purposes, the relevant operation is straightforward: given a prompt (the "context"), the LLM outputs a probability distribution over candidate next tokens, from which a token is sampled and appended to the context, with the process iterating until a stop condition is reached. Sampling temperature and top-p parameters control how concentrated the distribution is on its mode, with temperature near zero producing approximately deterministic output and higher temperatures producing more variable output.

For synthetic CV, the prompt has three parts. A persona block describes the respondent's demographic and (optionally) belief characteristics in natural language. A scenario block describes the policy and its randomized treatment attributes. A question block presents the referendum question with the randomized bid and prompts the LLM for a structured response. The LLM's response is then parsed for the binary support/oppose vote, which becomes the synthetic analog of y_i in Eq. 2. We describe our specific prompt structure, parsing logic, and validation procedures in Section 4.

It is worth stating plainly what this generation process is and is not. It is not a model of human preferences. The LLM has no utility function, incurs no cost when the policy passes, and

[2] A "token" in an LLM context is the fundamental unit of data–such as a word, part of a word, punctuation, or character–that large language models (LLMs) process and generate. In English, this is about four characters.

has no preference over policy outcomes that the bid amount could trade against. Its response to a CV prompt is whatever distribution over support and oppose tokens its training has induced, conditional on the persona, scenario, and question text. The empirical question we ask in this paper is whether that distribution—generated by a process structurally unlike the deliberation a human respondent engages in—happens to produce response patterns that approximate the human-survey patterns closely enough to support welfare analysis. There is no a priori reason to expect that it will, and the findings in Section 5 should be read as evidence on what currently does and does not transfer.

One feature of the generation process bears specifically on our methodology. The LLM's response is path-dependent on the preceding context, so the order in which persona elements are presented, the wording of the framing instructions, and the choice of whether to ask for a free-text rationale before or after the binary vote are all design decisions that can affect the response distribution. We address this through a surface-form sensitivity analysis reported in Appendix F, finding that surface-form variation produces shifts an order of magnitude smaller than across-LLM variation.

3 Data

We replicate two published contingent valuation studies. The studies span substantively different policy domains (energy versus forest pest control), sampling frames (nationally representative versus state-level opt-in panel), and elicitation formats (single-bounded referendum versus bounded and random-cost choice experiments), and provide a cross-context comparison of synthetic CV against published WTP benchmarks. Both studies are widely cited within the CV methodology literature, both report headline WTP estimates with published confidence intervals, and both authors made individual-level data available to us for the purposes of this replication.

We use only the variables collected by the original studies; we do not append any auxiliary data, and we do not modify the original randomization, treatment assignments, or sample composition. Synthetic personas (Section 4.1) are constructed one-to-one from original respondents, preserving each respondent's demographic profile, belief responses, treatment arm, and bid amount. This design choice—using the original sample as the reference population for the synthetic experiment, rather than constructing synthetic respondents from a separate demographic frame—ensures that any deviation between synthetic and empirical responses is attributable to the LLM's response process rather than to differences in sample composition. Summary statistics for both studies are reported in Table 1; full study design details, including verbatim CV question wording and the complete list of survey items used as belief inputs, are reported in Appendix A and Appendix B.

3.1 Study 1: National Clean Energy Standard

Aldy, Kotchen, and Leiserowitz (2012) elicit support for a hypothetical U.S. National Clean Energy Standard requiring electric power companies to obtain 80% of their energy from clean sources by 2035, conditional on a randomized increase in the respondent's annual electricity bill. The study uses a nationally representative online survey of 1,010 U.S. adults fielded between April 23 and May 12, 2011 via Ipsos KnowledgePanel (formerly known as Knowledge Networks). Two factors are independently randomized: the bid amount (eight levels from \$5 to \$155) and the technology mix that counts as "clean" (renewables only, renewables plus natural gas, or renewables plus nuclear). The original authors test for and fail to reject pooling across technology arms, and report a pooled binary logit specification as their preferred estimate. Mean WTP from the pooled specification is \$162 per household per year (95% CI: \$128–\$260).

Table 1: Summary of replicated studies.

	Study 1: NCES Aldy, Kotchen, and Leiserowitz (2012)	Study 2: HWA Giguere, Moore, and Whitehead (2020)
Sample		
Field period	April–May 2011	September 2017
Mode	Online survey	Online survey
Sampling frame	Nationally representative	North Carolina opt-in panel
Final N	1,010	907
Panel provider	Ipsos KnowledgePanel	SurveyMonkey
Policy and elicitation		
Policy	80% clean electricity by 2035	HWA control on public forest
Elicitation format	Single-bounded referendum	Bounded + random-cost choice
Cost framing	Annual electricity bill	Annual household tax (3 yrs)
Bid levels ($)	5, 25, 45, 65, 85, 105, 135, 155	25, 50, 100, 150, 200, 250[a]
Treatment arms		
Bid randomization	Yes (unequal probabilities[b])	Yes
Other randomized attributes	Clean-energy technology mix[c]	Eco. acres, social acres, treatment[d]
Survey content		
Demographic items	24	8
Belief / attitudinal items used	13	16
Headline WTP estimate		
WTP ($/household/year)	162[e]	56.53[f]
Original 95% CI	[128, 260]	[24, 75]
Original specification	Pooled binary logit	Conditional logit (ANA-adjusted)[g]

[a] The random-cost design uses bids of $50, $100, $150, and $200; the bounded design adds $25 and $250 levels.
[b] The four middle bids ($45–$105) were assigned with twice the probability of the two lowest and two highest bids.
[c] Three arms: renewables only; renewables + natural gas; renewables + nuclear.
[d] Acreage attributes vary across $\{2{,}500; 5{,}000; 7{,}500; 10{,}000\}$ acres for both ecologically and socially important categories; treatment method varies between chemical insecticide and biological control.
[e] Mean WTP from the pooled logit specification.
[f] Marginal WTP for an additional acre of "socially important" forest treated, at the ANA-adjusted conditional logit's preferred specification.
[g] ANA = attribute nonattendance, the focus of the original paper.

For our synthetic replication, we use the 24 demographic items collected by the original survey (e.g., age, gender, race/ethnicity, education, household income, urbanicity, employment status, marital status, household size, and political party affiliation) and 13 belief and attitudinal items covering climate change knowledge, trust in science, and political priority beliefs. The complete item list, including verbatim wording, is reported in Appendix B. The bid level and technology arm assigned to each original respondent are preserved exactly in the corresponding synthetic prompt.

3.2 Study 2: Hemlock Woolly Adelgid Control

Giguere, Moore, and Whitehead (2020) elicit support for a hypothetical North Carolina program to control hemlock woolly adelgid—an invasive insect that has caused substantial mortality in eastern hemlock stands across the Southern Appalachian region—through a randomized annual household tax over three years. The study uses an online opt-in panel of 907 North Carolina residents fielded in September 2017 via SurveyMonkey. The choice experiment varies four attributes: ecologically important acreage treated, socially important acreage treated, treatment method (chemical insecticide or biological control with predatory beetles), and annual household cost. Respondents are assigned to one of two formats: a *random-cost repeated* choice format in which bid amounts are drawn independently per question, or a *bounded* choice format in which bid amounts follow an ascending or descending sequence within respondent. The bounded format provides internal consistency checks and supports tests for attribute nonattendance, which is the focus of the original paper. The authors report a conditional logit with attribute-nonattendance adjustment as their preferred specification, yielding a marginal WTP of $56.53 per household per year for an additional acre of "socially important" forest treated (95% CI: $24–$75).

For our synthetic replication, we use the eight demographic items collected by the original survey (ZIP code, birth year, gender, marital status, presence of children under 18, race/ethnicity, education level, and 2017 personal income) and 16 belief and attitudinal items covering hemlock awareness, prior visits to affected forests, beliefs about treatment effectiveness, ecological values, and attitudes toward public funding of forest protection (Appendix B). Treatment arm assignments and bid amounts are preserved from the original respondent records.

3.3 Why These Two Studies

Three considerations drove case selection. First, both studies report transparent, reproducible specifications with sufficient detail to allow exact replication of the econometric analysis, which is necessary to attribute deviations between synthetic and empirical estimates to the LLM rather than to specification differences. Second, the headline WTP estimates lie within the range of plausible regulatory-relevance values—neither so small as to be dominated by sampling noise nor so large as to suggest pathological hypothetical bias—which makes the comparison to synthetic estimates substantively meaningful. Third, the studies differ in dimensions that allow us to assess generalization across published CV practice: Aldy, Kotchen, and Leiserowitz (2012) is nationally representative on a national policy with high media salience, and uses a single-bounded referendum; Giguere, Moore, and Whitehead (2020) is a state-level study on a regionally salient policy and uses bounded and random-cost choice formats. If synthetic CV behaves substantially differently across these two configurations, the contrast between the studies should reveal it.

We acknowledge two limitations of this case selection that we discuss further in Section 6.4. First, both studies are U.S. environmental CV with referendum-style or referendum-equivalent elicitation; we do not test synthetic CV in non-U.S. contexts, non-environmental domains, or non-referendum elicitation formats. Second, both studies' headline WTP estimates have appeared in subsequent publications and could plausibly enter LLM training data. We discuss the implications of this in Appendix H.

3.4 Data Access and Replication

Individual-level response data from both studies were obtained from the corresponding authors via direct request. We thank the original authors for sharing the data necessary to conduct this replication.

The synthetic responses generated in this paper, the persona templates and prompt structures used to generate them, the model configurations and API call records, and the analysis code are publicly archived at [**Zenodo / Dataverse DOI**]. The original individual-level human-survey data are not redistributed in our archive; readers wishing to verify our replication of the original logit specifications should request the original data from the original authors.

4 Methods

Our objective is to evaluate whether LLM-generated responses can substitute for human responses in referendum-style contingent valuation. To do so, we replicate two published CV studies—Aldy, Kotchen, and Leiserowitz (2012) on a U.S. National Clean Energy Standard and Giguere, Moore, and Whitehead (2020) on hemlock woolly adelgid control in North Carolina forests—using LLM-generated responses in place of the original human responses, while preserving each study's original demographic composition, experimental design, randomization structure, and econometric specification. This section describes the persona construction, prompt design, model selection, simulation protocol, and analytic procedures common to both replications. Study-specific details (bid distributions, treatment arms, choice formats) are summarized in Section 3 and reported in full in Appendix A.

4.1 Synthetic Persona Construction

For each respondent in the original study, we construct a synthetic persona that encodes the respondent's individual characteristics in natural-language form for use as input to an LLM. Each persona consists of three blocks: (i) a fixed system-prompt preamble specifying the simulation context, (ii) a demographic block enumerating the respondent's sociodemographic attributes, and (iii) an optional belief block enumerating the respondent's responses to non-CV survey items (attitudinal, knowledge, or experience questions).

Personas are generated programmatically from the original study's individual-level data. For respondent i with demographic vector $\mathbf{d}_i$ and belief vector $\mathbf{b}_i$, the persona P_i is constructed by template substitution: each attribute in $\mathbf{d}_i$ and (optionally) $\mathbf{b}_i$ is inserted into a fixed natural-language template at a designated slot. The template is identical across respondents within a study; only the slot values vary (see Table F.3.

We construct three classes of personas per respondent, corresponding to three information conditions. The *demographics-only* condition includes only the demographic block. The *demographics + all beliefs* condition appends the full battery of belief items. The *demographics + one belief at a time* condition appends each belief item individually in separate runs, isolating the marginal contribution of each belief item to synthetic accuracy. All main-text results use a single frozen production template (`v6.3`). Earlier versions of the template, the substantive design changes between versions (e.g., temporal anchoring, belief-block formatting), and per-version approval shifts are documented in Appendix Table F.3.

4.2 Contingent Valuation Prompts

After constructing the persona, we present the LLM with a CV question that exactly reproduces the wording, treatment assignments, and bid amounts of the original study. For respondent i, treatment arm t_i, and bid b_i, the CV prompt $Q_i(t_i, b_i)$ is constructed by inserting the original randomized treatment description and bid amount into the original study's question template. For Study 1, this is the referendum-style support/oppose question over the NCES policy with the randomized clean-energy technology mix and bid amount. For Study 2, this is the bounded or random-cost choice format question over the HWA treatment program with the randomized acreage and treatment-method attributes.

We preserve the exact treatment assignment and bid amount from the original respondent's record. That is, if respondent i in the original study was randomized to receive treatment arm $t_i =$ "natural gas + renewables" at bid $b_i = \$65$, the synthetic respondent constructed from i's persona is asked the same referendum question with the same treatment and bid. This design preserves the original study's experimental balance and randomization, so that any deviation between synthetic and empirical responses cannot be attributed to differences in treatment assignment.

The CV prompt instructs the LLM to return a structured response containing (i) a binary support/oppose vote, (ii) a confidence rating, and (iii) a free-text rationale. We use only the binary vote in our main analysis.

Question framing variation. As a robustness check, we run a parallel set of simulations in which the order of the response options is reversed ("oppose / support" versus "support / oppose"). This tests for ordering bias in LLM responses—a known concern in survey methodology that may operate differently in LLMs than in humans. We find no meaningful effect from this variation (Appendix E) and pool across orderings in our main results.

4.3 Model Selection

We evaluate ten LLMs spanning six providers, selected to represent (i) the current frontier of widely accessible models as of [**simulation date, e.g., October 2025**], (ii) variation along the cost-capability frontier from frontier-tier to lightweight models, and (iii) variation in training corpora and post-training procedures across providers. The full list is reported in Table 4.3.

Table 2: LLMs evaluated.

Provider	Model	Version / Release	Tier
OpenAI	GPT-5	[**XXXX-XX-XX**]	Frontier
OpenAI	GPT-5-mini	[**XXXX-XX-XX**]	Mid
Google	Gemini 2.5 Flash	[**XXXX-XX-XX**]	Mid
Google	Gemini 2.5 Flash-Lite	[**XXXX-XX-XX**]	Light
DeepSeek	DeepSeek-V3 (Chat)	[**XXXX-XX-XX**]	Frontier
DeepSeek	DeepSeek-R1	[**XXXX-XX-XX**]	Reasoning
Mistral	Mistral Medium 3.1	[**XXXX-XX-XX**]	Mid
Mistral	Mistral Small 3.2 (24B-Instruct)	[**XXXX-XX-XX**]	Light
Moonshot	Kimi-K2	[**XXXX-XX-XX**]	Frontier
Meta	Llama-4 Scout	[**XXXX-XX-XX**]	Mid

Notes: "Tier" is approximate and reflects each model's positioning within its provider's lineup at the time of simulation. All API calls were made between [**start date**] and [**end date**]. Specific model identifiers (e.g., `gpt-5-mini-2025-XX-XX`) are reported in Appendix D for replicability.

For each model we use the provider's chat-completion API with default sampling parameters except for the following standardizations applied across all models: temperature [**T**], top-p [**p**], and maximum response tokens [**XXX**]. Where a provider exposes a separate "reasoning" mode (e.g., DeepSeek-R1, GPT-5), we use the default reasoning configuration and do not pass the reasoning trace into our analytic dataset. We do not use system-level instruction tuning beyond the persona prompt described in Section 4.1.

4.4 Simulation Platform and Workflow

All simulations were executed on the Alethic-ISM (Instruction State Machine) platform, an open-source research workbench developed by two of the authors for composing and executing computational graphs with versioned, immutable states.[3] The platform encodes each experimental cell—a (study, information condition, model, run) tuple—as a node in a directed acyclic graph with edges representing transformations (LLM calls, data manipulations, validation checks). For our purposes the relevant property is that every synthetic response is linked back to its persona, its CV prompt, the model and parameters used to generate it, and the timestamp of generation, which supports full audit and replication of the simulation pipeline.

Each node in the graph applies an instruction to an input state via a processor, producing a new versioned, immutable output state. States are append-only and content-addressed; lineage is embedded in the data itself rather than maintained as a separate audit log, so the data record is the audit trail. Nothing is overwritten. This property matters for a study of this scale: with [**XXX,XXX**] total API calls across ten models, two studies, and multiple repetitions per cell, any ambiguity about which model version, prompt template, or parameter setting produced a given response would undermine replication. The platform resolves this by making every output state a permanent, queryable artifact linked to its full provenance chain.

The full execution graph for our experiment is summarized in Figure 1. The graph implements the cross-join $\mathcal{R} \times \mathcal{C} \times \mathcal{M}$ between respondents $\mathcal{R}$, conditions $\mathcal{C}$, and models $\mathcal{M}$, with separate downstream branches for the demographics-only, all-beliefs, and one-belief-at-a-time conditions. A key property of the graph architecture is that all ten models receive identical input states in parallel, each producing a separate immutable output state, which ensures that cross-model comparisons reflect differences in LLM response behavior rather than differences in input construction or sequencing. For Study 1 with $|\mathcal{R}| = 1{,}010$, $|\mathcal{M}| = 10$, and the one-belief-at-a-time condition adding $K = 9$ subcells, the cross-joined experiment evaluates [$\mathbf{1{,}010 \times 11 \times 10 = 111{,}100}$] persona–prompt–model combinations per repetition.

Figure 1: Alethic-ISM execution graph for the synthetic CV experiment. Each node represents a versioned state; each edge represents an instruction (LLM call, data join, or validation step). The complete graph encodes persona generation, prompt assembly, parallel LLM execution across all ten models, response parsing, and structured-output assembly for downstream econometric analysis.

4.5 Validation, Retries, and Output Structure

LLM responses are parsed against a structured schema requiring fields for the binary vote, confidence rating, and rationale. Responses that fail schema validation—for example, by returning ambiguous

[3] `https://github.com/alethicresearch/alethic-ism`. The platform is released under [**license**] and is not commercialized.

votes, refusing to answer, or producing malformed output—are retried up to three additional times with the same prompt. Persistent failures after four total attempts are coded as missing and excluded from the analytic sample. Schema-validation failure rates by model are reported in Appendix Table D; the highest failure rate across models was [**X.X%**] for [**model name**], and the resulting analytic samples preserve [**XX.X%**] or more of the original respondent count for every (model, condition) cell.

For each (respondent, condition, model, run) cell, the platform produces a structured output record containing: respondent ID, persona version hash, full prompt text, model identifier, sampling parameters, the parsed vote, confidence rating, rationale, raw API response, and timestamp. These records are exported as a flat panel for econometric analysis.

4.6 Repetition and Run Structure

Because LLM responses are stochastic at nonzero temperature, we run each (respondent, condition, model) cell multiple times to assess within-cell variability. For Study 1, we ran [$\mathbf{N}_1$] independent repetitions per cell; for Study 2, we ran [$\mathbf{N}_2$] repetitions. Each repetition uses an independent random seed where the provider's API supports seed control; for providers that do not, we rely on the default sampling stochasticity. The total number of API calls is reported in Table 4.6.

Table 3: Computational scope of the synthetic CV experiment.

	Study 1 (NCES)	Study 2 (HWA)
Respondents in original study	1,010	907
Information conditions	11	[**XX**]
Demographics only	1	1
Demographics + all beliefs	1	1
Demographics + one belief at a time	9	[**XX**]
LLMs evaluated	10	10
Repetitions per cell	[$\mathbf{N}_1$]	[$\mathbf{N}_2$]
Question-framing orderings	2	1
Total LLM API calls	[**XXX,XXX**]	[**XXX,XXX**]
Total synthetic responses generated	[**XXX,XXX**]	[**XXX,XXX**]

Notes: Total API calls aggregate across all (respondent, condition, model, repetition, ordering) cells, including failed-validation retries. Total synthetic responses count only schema-validated, non-missing observations entering the analytic sample.

4.7 Econometric Estimation

For each (study, model, condition) cell we estimate the same logit specification used in the original study. For Study 1, this is the pooled binary logit reported in Aldy, Kotchen, and Leiserowitz (2012), with the support/oppose vote as the dependent variable and bid amount, treatment indicators, and demographic controls as regressors. For Study 2, this is the conditional logit with attribute-nonattendance adjustment reported in Giguere, Moore, and Whitehead (2020), with the binary support vote, bid amount, ecologically- and socially-important acreage attributes, treatment-method indicator, and respondent-level controls. In both cases we use the original authors' preferred specification rather than re-optimizing; this preserves comparability between synthetic and empirical estimates.

Mean WTP is recovered as $\widehat{\text{WTP}} = -\hat{\alpha}/\hat{\beta}_b$ (or, for Study 2, marginal WTP for an additional acre of socially important forest, computed analogously from the relevant attribute coefficient and the bid coefficient). Confidence intervals around $\widehat{\text{WTP}}$ are constructed via the Krinsky-Robb parametric bootstrap with $B = [\mathbf{1{,}000}]$ replications, drawing from the asymptotic distribution of the logit coefficients.

For repetition-pooled estimates within a (study, model, condition) cell, we stack synthetic responses across repetitions and cluster standard errors at the respondent level. For the equally-weighted ensemble (the "Aggregated" results in Section 5), we pool synthetic responses across all ten models and re-estimate the same logit specification on the pooled panel.

4.8 Robustness and Limitations

We pre-specify three classes of robustness checks: (i) temporal contamination, (ii) prompt sensitivity, and (iii) persona-version sensitivity. Results are reported in Appendix F.

Temporal contamination. The Study 1 data were collected in 2011 and the Study 2 data in 2017, both well before the training cutoff of any model in our sample. To assess whether LLM responses are influenced by post-collection-window information (e.g., subsequent climate policy developments, post-2017 forest management literature), we run a held-out subsample under a stricter persona prompt that explicitly instructs the model to ignore any information acquired after the original survey window, and compare results.

Prompt sensitivity. We re-run a subsample of cells with [**X**] alternative persona templates that vary surface features (sentence ordering, phrasing of the framing instruction, formatting of the demographic block) while preserving substantive content. Cell-level mean differences are reported in Appendix Table F.2.

Persona-version sensitivity. The persona templates evolved over the course of the project as we refined wording and structure. We report the version identifier of the production template used for each study and provide cross-version comparisons for cells run under multiple template versions in Appendix Table F.3.

Limitations. Three limitations are inherent to the design. First, our synthetic respondents are constructed from the original studies' respondent pools, so any non-representativeness in the original samples (e.g., online-panel coverage in Giguere, Moore, and Whitehead (2020)) is preserved in the synthetic sample. Second, we evaluate models available at a particular point in time; capability differences across model generations may render specific quantitative findings outdated, though we expect the qualitative patterns—heterogeneity across models, slope flatness, sensitivity to belief inputs—to persist. Third, we cannot rule out that any particular LLM's training data included direct exposure to the original studies or their published results. We discuss this last concern in Section 6 and report a leakage-detection probe in Appendix H.

4.9 Reproducibility

Full Alethic-ISM workflow graphs, persona templates, prompt templates, model parameters, and analysis code are available at [**repository URL**]. Per-respondent synthetic response data are available at [**data repository URL**], subject to the data-use restrictions of the original studies.

5 Results

We present results in three subsections. Section 5.1 reports the synthetic replication of Aldy, Kotchen, and Leiserowitz (2012); Section 5.2 reports the synthetic replication of Giguere, Moore, and Whitehead (2020); and Section 5.3 synthesizes patterns across studies. Throughout, we evaluate synthetic responses along three dimensions: (i) reproduction of the referendum response distribution at each bid level, (ii) recovery of mean willingness to pay (WTP) relative to the original logit estimate, and (iii) sensitivity to the inclusion of belief variables in the synthetic respondent profile.

5.1 Study 1: U.S. National Clean Energy Standard

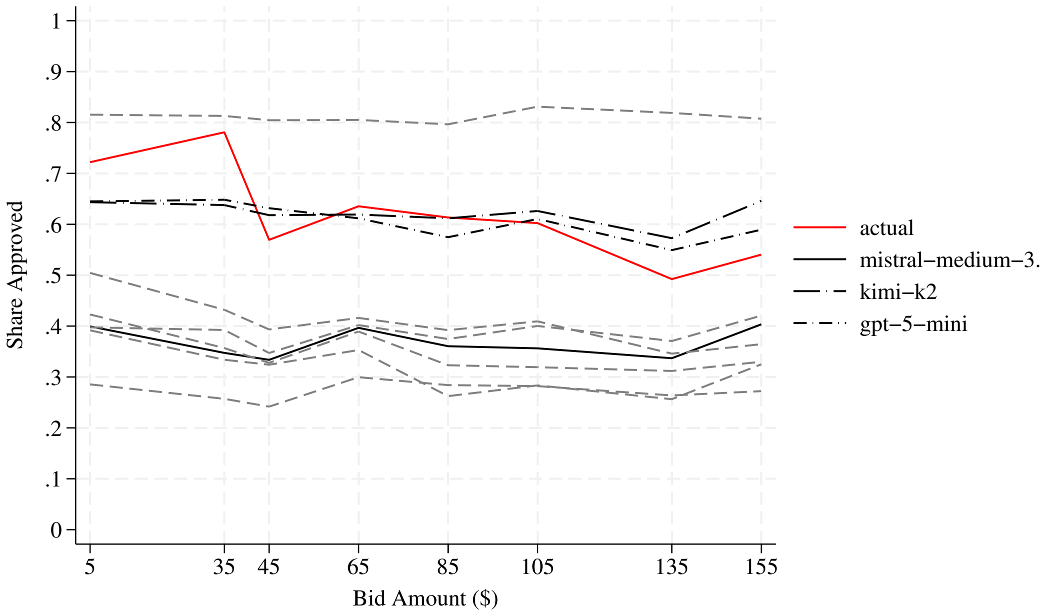
Aldy, Kotchen, and Leiserowitz (2012) estimate a mean WTP of \$162 per household per year (95% CI: \$128–\$260) for an 80% clean electricity standard by 2035, based on a pooled logit over 1,010 nationally representative respondents at bids ranging from \$5 to \$155. The empirical share-approving curve declines from approximately 0.72 at the \$5 bid to 0.54 at the \$155 bid, with a marked drop between the \$35 and \$45 bid points. We compare synthetic responses to this benchmark in three information conditions: demographics-only, demographics + all beliefs, and demographics + one belief at a time.

When synthetic respondents are constructed using only sociodemographic attributes from the original survey, most LLMs substantially underestimate the share approving the policy at every bid level. Figure 2 plots synthetic approval shares against the empirical curve for all ten models. Seven of the ten produce approval shares clustered between 0.25 and 0.45, well below the empirical curve. Two models—Kimi-K2 and GPT-5-mini—track the empirical approval curve closely, with mean absolute deviations of [**X.XX**] and [**X.XX**] percentage points across bid levels, respectively. Across all models, including the two that track the empirical level well, the synthetic approval-bid curves are visibly flatter than the empirical curve—a pattern we return to when we discuss WTP and again in Section 5.3, where it appears across both studies.

Appending the full battery of belief and attitudinal responses to the synthetic persona produces a substantial level shift in approval shares for most models (Figure 3). The previously underperforming models—including Mistral-Medium-3.1, DeepSeek-R1, Gemini-2.5-Flash, and Gemini-2.5-Flash-Lite—shift upward and now bracket the empirical curve. Notably, Kimi-K2 and GPT-5-mini, which performed best without beliefs, do *not* improve when beliefs are added; in some bid ranges their fit deteriorates. The slope of the synthetic approval-bid curve remains shallower than the empirical slope across all models, indicating that even with belief information, LLM respondents are systematically less price-sensitive than human respondents. Belief inputs in this study appear to operate primarily on the level of approval rather than on the slope.

Figure 4 summarizes the implied mean WTP from logit estimates fit to synthetic responses, with the \$162 benchmark marked as a vertical reference line. The pattern of WTP estimates across LLMs and information conditions is best understood through the lens of how each model fares jointly on the level and slope of its approval-bid curve, in light of the WTP formula $\widehat{\mathrm{WTP}} = -\hat{\alpha}/\hat{\beta}_b$ established in Eq. 3. Three patterns at this joint level-and-slope view emerge.

Figure 2: Approval share by bid amount, NCES (demographics-only condition)



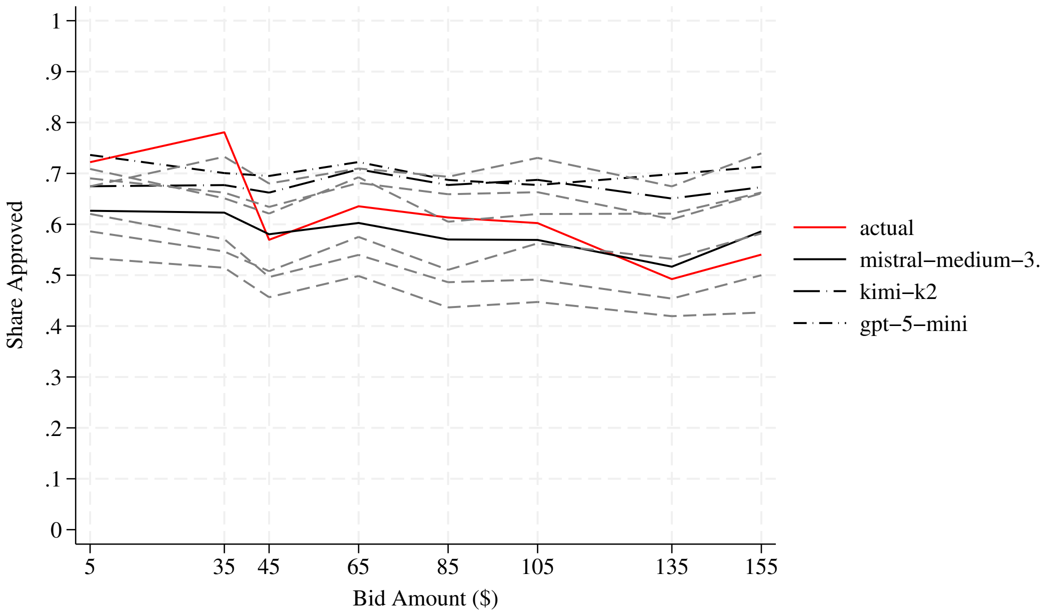
Note: Solid red line plots the empirical share approving from Aldy, Kotchen, and Leiserowitz (2012); dashed lines plot synthetic approval shares for each of the ten LLMs evaluated.

First, in the demographics-only condition, WTP estimates are negative or near zero for seven of the ten models, with DeepSeek-Chat-V3.1 producing an extreme negative estimate of approximately -$843. These seven models are mis-calibrated on both dimensions: their approval levels sit well below the empirical curve, and their slopes are flatter than empirical, with the level error dominating to push WTP negative. Demographics alone, for these models, provide neither a useful prior on the level of policy support nor on its sensitivity to cost. The two exceptions are Kimi-K2 and GPT-5-mini, whose demographics-only WTP estimates fall within a plausible range of the benchmark. These are the models for which level fit is reasonable and slope flatness is mild enough that the WTP ratio remains in a sensible range. Kimi-K2 in the demographics-only condition is the closest the experiment gets to an existence proof that current synthetic CV can recover a published human-survey benchmark from sociodemographic inputs alone, and we treat it as such.

Second, adding beliefs shifts most models' WTP estimates from negative to positive territory, but the magnitude of this update varies dramatically across LLMs—from modest revisions of under $200 (DeepSeek-R1) to revisions exceeding $[**1700**] (Llama-4-Scout). Because beliefs in this study shift levels without straightening slopes, the resulting WTP estimates are arithmetic products of a (now plausible) numerator and a (still flat) denominator, which tends to push the ratio above the empirical benchmark. Several models that produce sensible-looking approval-bid curves with beliefs nonetheless overshoot the $162 benchmark by a wide margin for this reason.

Third, the simple equally-weighted aggregate across all ten LLMs does not outperform the best individual model in either condition: the aggregated WTP without beliefs is -$9 (95% CI: [-$17, -$2])

Figure 3: Approval share by bid amount, NCES (demographics + all beliefs condition)



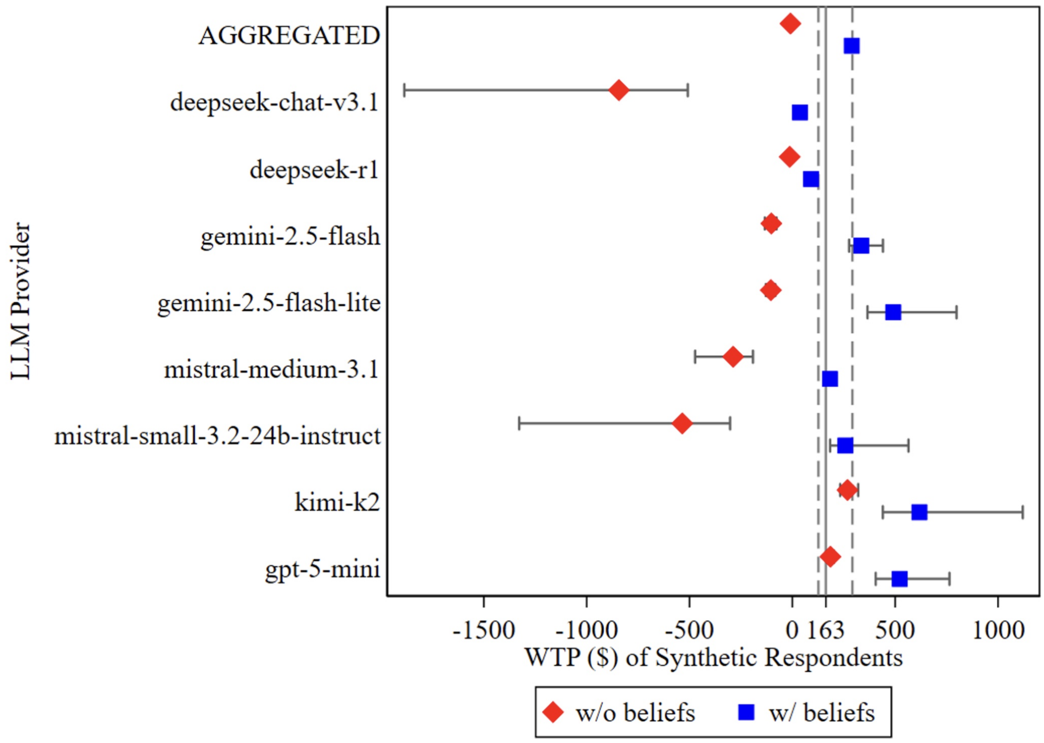
Note: Solid red line plots the empirical share approving from Aldy, Kotchen, and Leiserowitz (2012); dashed lines plot synthetic approval shares for each of the ten LLMs evaluated.

and with beliefs is \$289 (95% CI: [\$268, \$314]), neither of which dominates Mistral-Medium-3.1's demographics+belief estimate of \$183. Aggregation reduces the variance contributed by individual models' idiosyncratic level errors but cannot offset the slope flatness that the underlying training distributions appear to share.

To isolate the marginal contribution of each belief question, we re-ran the experiment appending each of the 13 belief questions individually (Q1–Q13). Figure 5 reports approval-bid curves for each single-belief specification. Two findings emerge. Adding a single well-chosen belief question often produces a closer fit to the empirical curve than adding the full belief battery, suggesting that some belief questions introduce noise rather than signal. Belief questions Q1 ("Do you think that global warming is happening?") and Q3 (whether scientists think global warming is happening) systematically degrade fit when included. And in this condition, Llama-4 Scout and GPT-5-mini consistently outperform other LLMs on response shares. Llama-4 Scout's case is particularly informative for the level-versus-slope point. Despite tracking individual approval shares well at every bid level—a level fit comparable to the best models in the experiment—its implied WTP estimates exceed those of other models by an order of magnitude. The reason is its near-flat bid-response slope: with $\hat{\beta}_b$ close to zero, the ratio $-\hat{\alpha}/\hat{\beta}_b$ explodes regardless of how well $\hat{\alpha}$ recovers empirical levels. Llama-4 Scout in this condition is the clearest single illustration in our experiment of a synthetic respondent that is well-calibrated on the response margin and badly mis-calibrated on the welfare margin.

Taken together, the Study 1 results illustrate a typology of synthetic respondents that we will see again in Study 2: models that recover levels but flatten slopes (Llama-4 Scout, the prototypical

Figure 4: Mean WTP estimates for the NCES policy by LLM



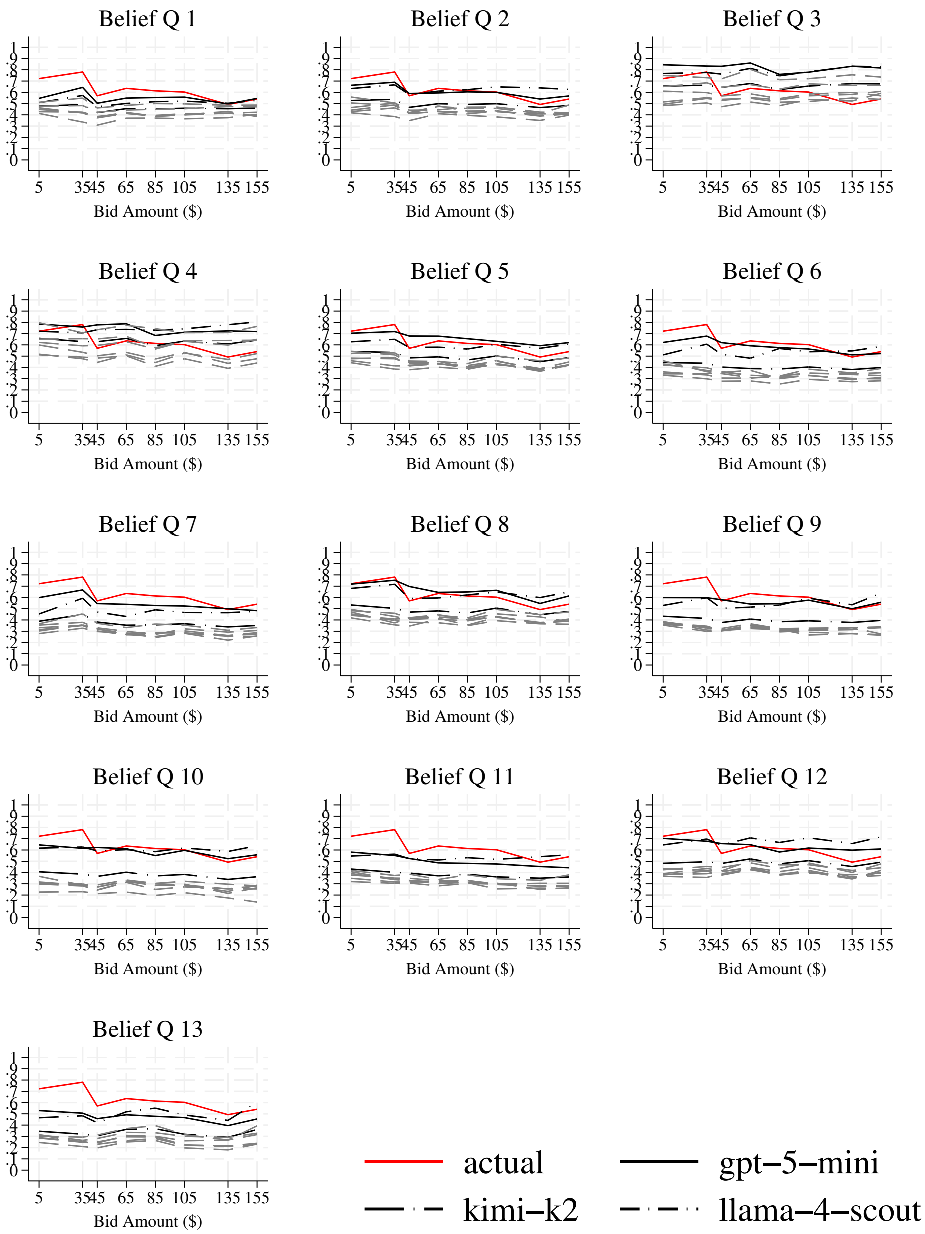
Note: Red diamonds: demographics-only condition. Blue squares: demographics + all beliefs condition. Vertical solid line marks the Aldy, Kotchen, and Leiserowitz (2012) benchmark of $162; dashed vertical lines mark the original 95% CI bounds.

"WTP-inflating" archetype), models that fail on both dimensions (most demographics-only configurations), and a small set of models for which levels and slopes happen to land close enough to empirical that synthetic WTP recovers the benchmark (Kimi-K2 in the demographics-only condition). The implied WTP from a logit fit to synthetic data depends jointly on the level and the slope of the synthetic approval-bid curve, and accuracy on the response margin does not imply accuracy on the welfare margin—a point that follows structurally from the WTP formula in Eq. 3 but is illustrated sharply by the cross-LLM variation in Figure 4.

5.2 Study 2: Hemlock Woolly Adelgid Control

Giguere, Moore, and Whitehead (2020) estimate a marginal WTP of $56.53 per household per year (95% CI: $24–$75) to treat a "socially important" acre of forest, based on a conditional logit over 907 North Carolina residents. The study employs two choice formats: a *bounded* choice experiment with cost increments from $25 to $250, and a *random-cost repeated* choice experiment with bids from $50 to $200. The empirical approval curve declines from approximately 0.66 at $50 to 0.45 at

Figure 5: Approval share by bid amount, NCES (demographics + one-belief-at-a-time condition)



Note: Each panel adds a single belief question (Q1–Q13) to the demographics-only persona. Solid red line: empirical share approving. Dashed lines: top-performing LLMs.

$150–$200 in the random-cost design, and from approximately 0.76 at $25 to 0.27 at $250 in the bounded design. The bounded design's wider bid range and steeper empirical slope make it a tighter test of synthetic slope recovery than the random-cost design.

In contrast to Study 1, several models track the empirical curve reasonably well in the bounded-cost experiment without belief inputs (Figure 6). Kimi-K2 and Llama-4 Scout produce approval-bid curves that closely follow the empirical curve in the bounded design. GPT-5-mini, by contrast, dramatically *over*-approves at every bid level, with synthetic approval shares above 0.75 even at the highest bids—the opposite failure mode from its Study 1 performance, and a useful illustration that level errors can run in either direction depending on the policy domain.

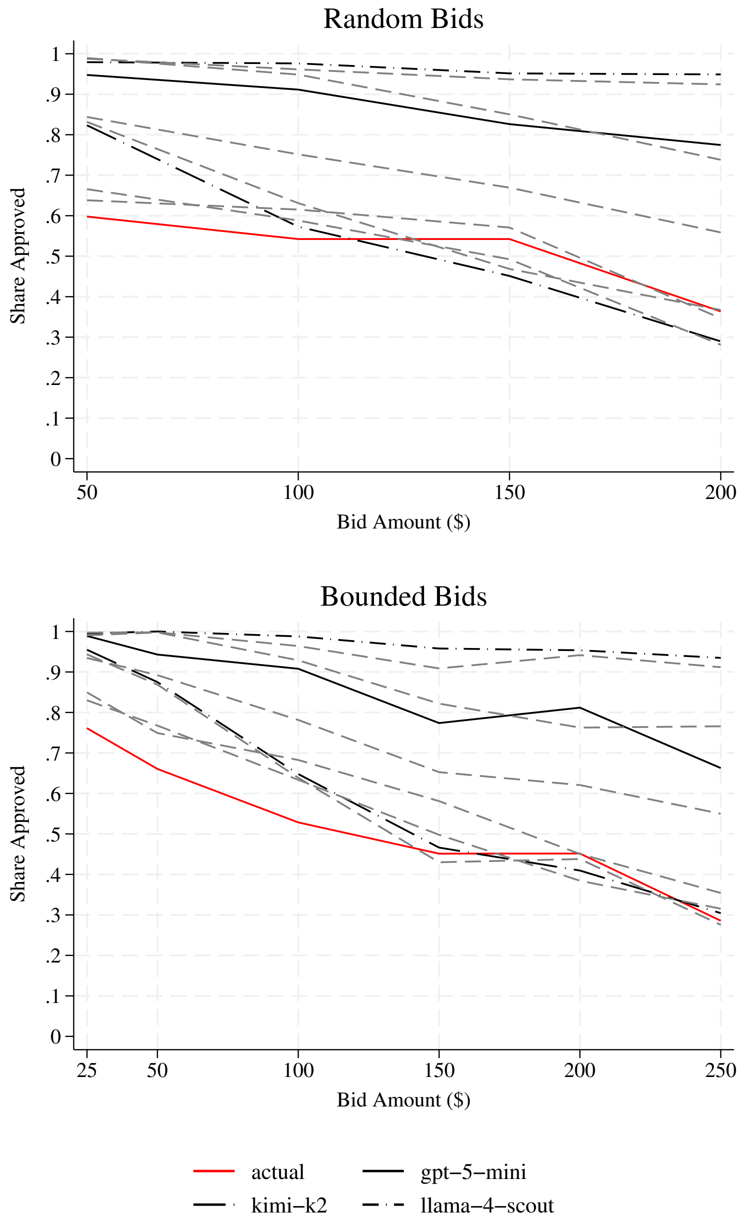
Adding beliefs produces mixed and somewhat counterintuitive effects (Figure 7). Several models that performed adequately without beliefs (notably Kimi-K2 and Llama-4 Scout) shift *downward*, now under-approving relative to the empirical curve. GPT-5-mini's over-approval problem is not corrected by the addition of beliefs. The bounded-cost experiment produces noticeably better fits than the random-cost experiment under both information conditions, particularly at moderate bid amounts.

Figure 8 reveals two notable departures from the Study 1 pattern that become more legible when read through the lens of levels and slopes. In the random-cost repeated choice experiment, WTP estimates without beliefs are clustered near or modestly above the $56.53 benchmark for most models, with the simple aggregate landing within the empirical 95% CI. This is the closest any cross-model average in our experiment comes to the human-survey benchmark, and the reason it works appears to be that the random-cost design's narrower bid range ($50–$200) makes moderate slope flatness less mechanically consequential than it is over a wider range: when the empirical bid coefficient is itself relatively shallow over the experiment's range, a synthetic respondent that is slightly flatter still produces a plausible WTP ratio. Adding beliefs to this configuration shifts several models substantially higher (e.g., Gemini-2.5-Flash to roughly $425) and produces wider confidence intervals overall, consistent with the Study 1 pattern that beliefs move levels without straightening slopes.

In the bounded choice experiment, the pattern reverses. WTP without beliefs is roughly centered on the benchmark for most models, but adding beliefs produces several large *negative* WTP estimates—most notably for Llama-4 Scout (approximately −$762) and Mistral-Medium-3.1 (approximately −$127). The aggregated estimate without beliefs in the bounded design overshoots the benchmark, while the aggregated estimate with beliefs lands negative. The bounded design's wider bid range makes both dimensions of synthetic mis-calibration more visible: small slope errors translate into larger WTP errors than in the random-cost design, and level shifts induced by belief inputs that move synthetic approval below 50% at the highest bids can flip the sign of the implied $\hat{\alpha}$, producing the observed negative WTP estimates. The combination of belief-induced level shifts with persistent slope flatness is what generates the more extreme WTP estimates in this design.

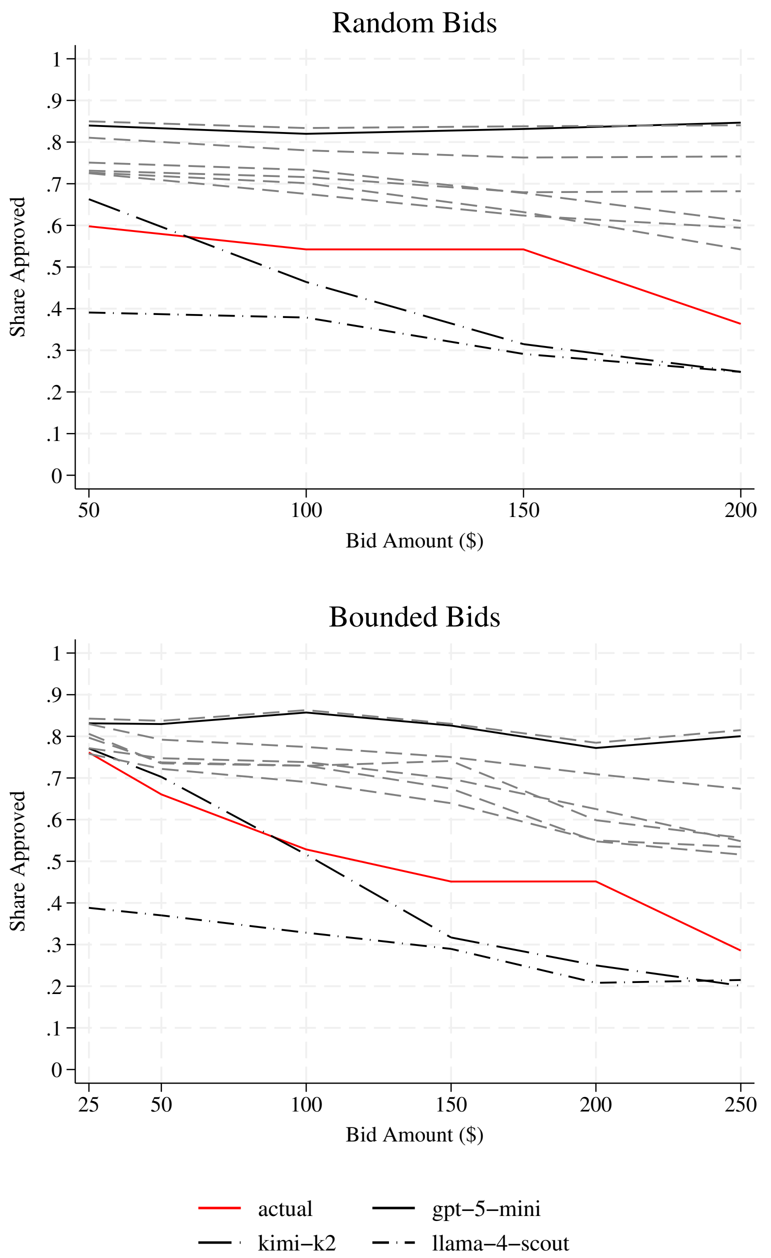
The Study 2 results map onto the typology of Study 1 with one important generalization. The same models that tracked levels well in Study 1 (Kimi-K2, GPT-5-mini, Llama-4 Scout) show different level performance in Study 2 (Kimi-K2 still good in bounded; GPT-5-mini badly over-approving; Llama-4 Scout moves down with beliefs into under-approval). The slope flatness, however, is consistent: across both choice formats and both information conditions, synthetic approval-bid curves are flatter than the empirical curves. Level performance is policy-domain-specific—a model that recovers the level of approval for clean energy policy may not recover it for forest pest control—while slope behavior appears to be model-specific and domain-invariant. This combination is what generates the within-study choice-format reversals visible in Figure 8: models whose level error happens to align with the empirical curve in one format produce reasonable WTP, while in the other format the same model's level error works against the empirical curve and the combination with

Figure 6: Approval share by bid amount, HWA (demographics-only condition)



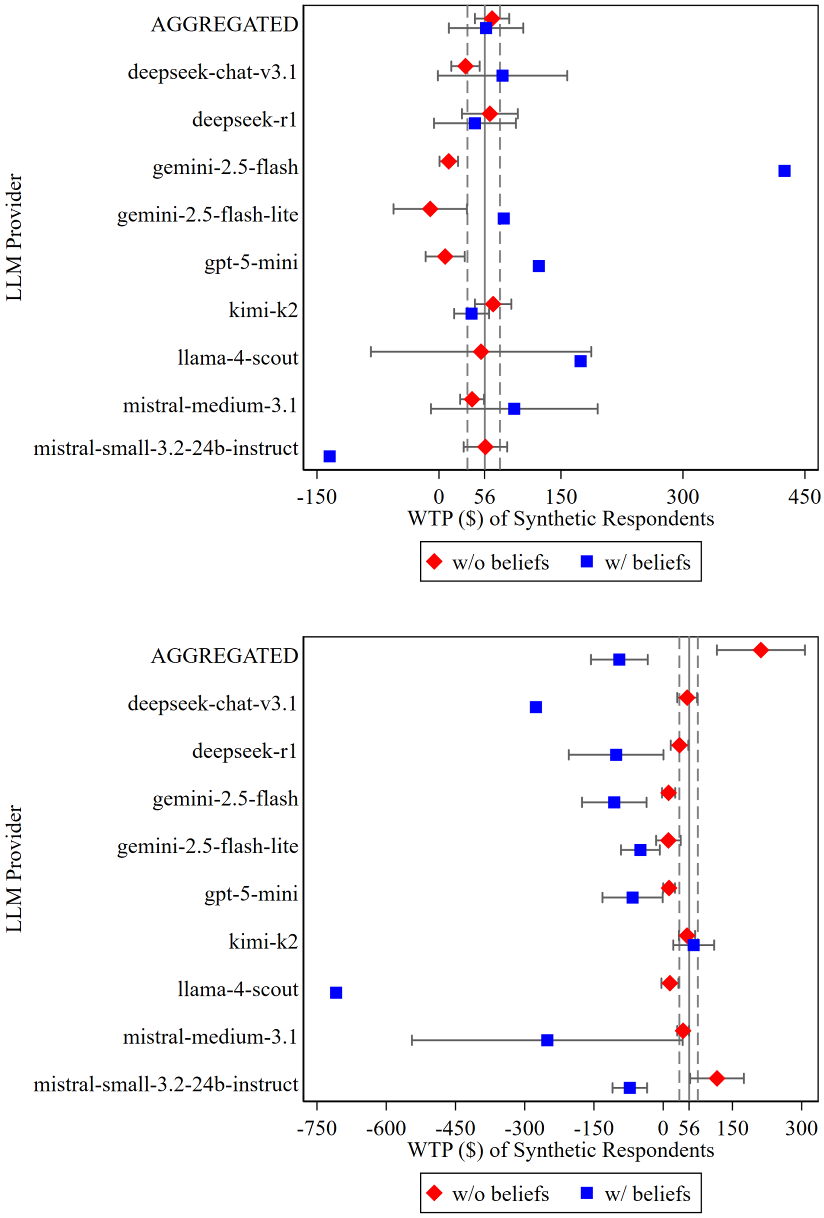
Note: Upper panel: random-cost repeated choice experiment. Bottom panel: bounded-cost choice experiment. Solid red line: empirical share approving. Dashed lines: synthetic approval shares for each LLM.

Figure 7: Approval share by bid amount, HWA, demographics + all beliefs condition)



Note: Upper panel: random-cost repeated choice experiment. Lower panel: bounded-cost choice experiment.

Figure 8: Mean WTP estimates for HWA policy by LLM



*Note: Left panel: random-cost repeated choice experiment. Right panel: bounded-cost choice experiment. Red diamonds: demographics-only. Blue squares: demographics + all beliefs. Vertical solid line marks the Giguere, Moore, and Whitehead (2020) benchmark of $56.53; dashed vertical lines mark the original 95% CI bounds. Estimates with confidence intervals exceeding ±$[**XXX**] are omitted for visual clarity; all estimates are reported in Table G.2.*

persistent slope flatness produces extreme WTP estimates. We discuss the practical implications of this combination in Section 6.

5.3 Cross-Study Patterns

Comparing across the two studies yields four robust patterns and one notable disanalogy.

Heterogeneity dominates. In both studies, the gap between the best- and worst-performing models on any given metric is large relative to the gap between with-beliefs and without-beliefs conditions for a fixed model. Mean absolute deviations across LLMs span [**X.XX**] to [**X.XX**] in Study 1 and [**X.XX**] to [**X.XX**] in Study 2. Practitioners should treat LLM selection as a first-order design decision.

Slope flatness is universal. Synthetic approval-bid curves are systematically flatter than the empirical curves across both studies, both choice formats, and almost every model. The empirical-relative slope ratios reported in Table 5.3 are quantitatively similar across the two studies, despite their different policy domains and bid ranges. We treat this as an observed regularity rather than a diagnosed mechanism: the pattern is consistent enough to merit naming, but its underlying source—training-distribution averaging across heterogeneous price effects, attenuated treatment of bid as a substantive cost, numerical-anchoring artifacts, or some combination—is not pinned down by the analysis we report here. Diagnosing the source is a natural priority for follow-up work. The mechanical implication for WTP estimation is, however, immediate: when synthetic slope is flatter than empirical slope and synthetic level approximates empirical level, synthetic WTP is inflated relative to the empirical benchmark, sometimes by an order of magnitude (cf. Llama-4 Scout in Section 5.1).

Aggregation does not rescue weak models. In both studies, the simple equally-weighted aggregate across all ten LLMs is dominated by the best individual model in WTP recovery. The intuition is straightforward: aggregation reduces variance but not bias, and the bias from slope flatness is correlated across models because it arises from features shared across the underlying training distributions.

Belief-question quality varies sharply. The one-belief-at-a-time results from Study 1 indicate that belief questions are not interchangeable: a few questions carry most of the predictive signal, while others actively degrade fit. Practitioners using LLMs to simulate CV experiments should pre-screen belief items rather than include the full survey battery by default.

Disanalogy: belief value is context-dependent. The most striking cross-study contrast concerns the marginal value of belief information. In Study 1, demographics alone produce mostly negative WTP estimates, and beliefs are essentially required to recover plausible levels. In Study 2, demographics alone produce reasonable WTP estimates in the random-cost design, and beliefs sometimes *worsen* fit. We conjecture that this reflects the relative density of the policy domain in the LLM training distribution: clean energy policy is heavily discussed in online text, so demographic priors may map cleanly onto stable approval patterns, but with substantial baseline mis-calibration that beliefs can correct. Hemlock forest protection is a localized issue where the LLM's prior may already be diffuse, making belief inputs both more informative and more easily corrupted by noisy items. Distinguishing these explanations is a priority for future work.

Table 4: Cross-study summary of synthetic CV performance.

Metric	Study 1 (NCES) Demo only	Study 1 (NCES) + Beliefs	Study 2 (HWA) Demo only	Study 2 (HWA) + Beliefs
Best-LLM mean abs. deviation (pp)	[X.XX]	[X.XX]	[X.XX]	[X.XX]
Aggregate WTP error ($)	[XXX]	[XXX]	[XXX]	[XXX]
Best-LLM WTP error ($)	[XXX]	[XXX]	[XXX]	[XXX]
Synthetic-to-empirical bid coefficient ratio	[0.XX]	[0.XX]	[0.XX]	[0.XX]
Share of LLMs with WTP in original 95% CI	[X/10]	[X/10]	[X/10]	[X/10]

Notes: Mean absolute deviation computed across bid levels and averaged across LLMs. Aggregate WTP refers to the equally-weighted ensemble across all ten models. The synthetic-to-empirical bid coefficient ratio is the cross-LLM mean of each model's bid coefficient divided by the corresponding empirical bid coefficient; values below one indicate flatter synthetic curves than empirical curves.

6 Discussion

Our results provide a structured assessment of whether large language models can substitute for human respondents in contingent valuation. The headline finding is that current LLMs cannot deliver point estimates of willingness to pay that match human-survey benchmarks under naive deployment, but the deviations follow a regular pattern that has direct implications for how synthetic CV should be evaluated and used.

6.1 What Synthetic CV Can and Cannot Do

Synthetic CV can reproduce qualitative referendum behavior. Across both studies, the best-performing LLMs reproduce the empirical share-approving curve to within [**X.X**] percentage points at every bid level under a properly specified persona. The qualitative shape of the bid-response relationship—monotonic decline in approval with bid amount, modest convexity—is recovered by most models in most conditions. For applications where the analyst needs only the direction or relative magnitude of policy support across stakeholder subgroups, current synthetic CV is already useful.

Synthetic CV can probe survey design before fielding. Because synthetic experiments cost approximately [**$X–$Y**] per [**1,000**] simulated responses and complete in [**hours**] rather than months, they can be used to stress-test wording, ordering, and treatment-arm specifications before committing to a costly human-subjects study. Our finding that surface-form prompt variation produces shifts an order of magnitude smaller than across-LLM variation (Appendix F) suggests that within-LLM piloting can identify gross design problems—ambiguous wording, leading framing, treatment arms that produce identical responses—without false signals from prompt sensitivity. Synthetic CV piloting is a clear, near-term, defensible use case.

Synthetic CV cannot deliver point estimates of WTP at face value. Even the best-performing models produce WTP estimates that diverge substantially from human-survey benchmarks under naive deployment. The slope-flatness pattern documented in Section 5 has a direct mathematical implication for WTP recovery: errors in the price coefficient $\hat{\beta}_b$ produce inversely proportional errors in $\widehat{\mathrm{WTP}} = -\hat{\alpha}/\hat{\beta}_b$, so a model that recovers the level of approval to within five percentage points but produces a slope half the empirical slope will inflate WTP by a factor of roughly two—and our synthetic-to-empirical slope ratios are typically lower than 0.5. Practitioners who use uncalibrated synthetic WTP for benefit-cost analysis in regulatory impact assessments will produce systematically biased estimates, and the direction of the bias will not be obvious without external validation.

What synthetic CV cannot yet do at all. Three features of human CV studies have no demonstrated synthetic analog. First, scope sensitivity tests—the foundation of validity assessment in modern CV (Johnston et al. 2017)—require synthetic respondents to update WTP in proportion to the magnitude of the good, and our preliminary evidence in Study 2 (Section 5.2) suggests LLMs largely fail this test even when their level fit is reasonable. Second, the cheap-talk and consequentiality scripts that mitigate hypothetical bias in human surveys have no obvious analog for an entity that does not bear costs and is not voting. Third, certainty follow-up questions—which screen for genuine versus casual support—produce response patterns from LLMs that we have not been able to interpret in a way that maps onto the human literature. Each of these is a candidate for future research, but none can currently substitute for the corresponding human-survey instrument.

6.2 Synthetic CV as Benefit Transfer

We argue that the most natural intellectual home for synthetic CV is the benefit transfer literature (Rosenberger and Loomis 2017; Loomis 2015; Boyle et al. 2010), not the primary CV literature, and we believe this reframing has both substantive and tactical implications.

Benefit transfer asks: when a regulator needs WTP for a policy in a context where no original study has been conducted, how should they extrapolate from existing studies in related contexts? Existing methods include unit value transfer (apply the WTP estimate from a similar study), function transfer (apply the WTP equation from a similar study, plugging in local covariates), and meta-analytic transfer (synthesize multiple studies into a transfer function). All of these methods rely on the assumption that preferences in the target context can be extrapolated from preferences observed in source contexts.

Synthetic CV is, viewed this way, a form of benefit transfer in which the source context is the universe of text on which the LLM was trained. The LLM acts as an implicit meta-analytic engine, having absorbed (an unknown subset of) the published CV literature plus large quantities of related social, political, and economic text, and producing a synthesized response when queried with a target-context persona. The slope-flatness pattern we document has a natural interpretation under this analogy: if the training distribution averages over many studies with heterogeneous price effects, the resulting bid coefficient should be attenuated toward the mean of that distribution, in the same way that meta-analytic transfer is known to produce attenuated transfer coefficients (Bergstrom and Taylor 2006). We do not claim our results demonstrate this is the operating mechanism—other accounts of the slope flatness are also consistent with our evidence—but the parallel suggests that the corrections developed in the benefit transfer literature, including generalizability tests, convergent validity checks, and transfer error metrics, should be brought to bear on synthetic CV.

This reframing has two implications. First, it places more reasonable expectations on synthetic CV: we should evaluate it against benefit transfer benchmarks (transfer errors of [**20–40%**] are common in the conventional literature), not against the standards of an original study. Second, it suggests a hybrid workflow in which synthetic CV is used to augment, rather than replace, benefit transfer: the LLM's persona-conditioning capacity allows for finer-grained transfer than meta-analytic functions estimated on study-level summaries, while the conventional benefit transfer literature provides validity criteria that pure synthetic studies lack. We see this as the most promising near-term direction for LLMs in environmental valuation: synthetic CV is plausibly a faster, more flexible benefit transfer engine, not a substitute for original CV studies.

A note on the scope of this analogy. The training-distribution-averaging account predicts slope flatness for general-purpose LLMs trained on broad text corpora. It does not predict slope flatness for specialized models that might in principle be trained on curated CV corpora with explicit grounding in price elasticity, demographic-conditioned WTP, and stated-preference elicitation conventions. Whether such specialized models are worth building, given the data-availability and validation challenges, is a separate question from whether current general-purpose LLMs are useful for synthetic CV—and it is a question that the field has not yet engaged seriously.

6.3 Implications for Regulatory Practice

Federal agencies producing regulatory impact analyses under Executive Order 12866 and OMB Circular A-4 face a recurring problem: the costs and benefits relevant to their analyses include non-market values that are expensive and slow to estimate using conventional CV, and the consequence of neglecting these values in the analysis is regulatory decisions made without full consideration of relevant costs and benefits.

Our results do not support the use of synthetic CV as a drop-in substitute for conventional CV in RIAs. The slope-flatness pattern produces biased point estimates of WTP that, if used in net-benefit calculations, will systematically distort regulatory choices. We do, however, see three uses for synthetic CV in regulatory practice that our results support.

First, synthetic CV is well-suited to *threshold analysis*, the practice recommended in Circular A-4 of identifying break-even values for non-quantified factors. A regulator who can establish that stakeholders' WTP for a non-market good plausibly exceeds some lower bound—and can do so quickly using synthetic CV—is in a stronger position to argue that the good warrants consideration in the analysis even without a precise point estimate.

Second, synthetic CV is well-suited to *sensitivity testing* of conventional benefit transfer. A regulator using a transferred WTP estimate from a published study can use synthetic CV with the target-context demographics to assess whether the transferred value is plausible given the local population's characteristics, and can flag cases where the transferred value appears systematically off.

Third, synthetic CV is well-suited to *pilot design* for cases where conventional CV will eventually be conducted. The cost and time of a full CV study is largely in the design and survey-administration phases; synthetic piloting can identify wording problems, treatment-arm collapse, and ordering effects in advance, reducing the risk that a costly study produces unusable results.

We do not believe synthetic CV is ready to substitute for conventional CV in cases where regulatory decisions hinge on precise WTP point estimates. We do believe it is ready, today, to expand the set of values that can be considered in regulatory analysis when the alternative is leaving them unexamined. The history of stated preference methods suggests that the field's most productive moves have come from expanding the range of values that can be brought into benefit-cost analysis while maintaining methodological discipline about the resulting estimates' reliability. Synthetic CV, used carefully, fits squarely in that tradition.

6.4 External Validity and Capability Evolution

Three external validity concerns deserve attention.

Generalization across CV contexts and elicitation formats. Our two studies span clean energy policy and forest pest control— substantively different domains, but both within U.S. environmental policy and both using referendum-style elicitation. We do not test synthetic CV on health valuation, mortality risk reduction, cultural heritage protection, animal welfare, or ecosystem services in developing-country contexts, all of which raise distinct issues. Nor do we test elicitation formats other than referendum and bounded/random-cost choice; open-ended WTP elicitation, payment cards, and ranking-based discrete choice experiments have different cognitive demands and may elicit different patterns from LLM respondents. The slope-flatness pattern we document appears across both of our studies and both choice formats in Study 2, but the level of bid sensitivity and the marginal value of belief inputs may differ across contexts in ways that our two studies cannot fully adjudicate.

Capability evolution. Our results reflect LLM capabilities as of [**simulation date**]. The history of LLM development since [**2020**] has been one of rapid capability gains on a broad range of tasks, and there is no obvious reason to expect synthetic CV to be exempt. Whether the slope-flatness pattern we document will abate as models improve is genuinely unclear, and depends in part on which kind of improvement is meant. General capability gains from continued pre-training and post-training of frontier models may reduce slope flatness as the treatment of numerical magnitudes becomes more sophisticated, but this is not assured: if slope flatness reflects training-distribution averaging across heterogeneous price effects, it may persist as long as broad-text corpora are the training substrate. A separate possibility is that future synthetic CV will rely on specialized models—trained on curated stated-preference corpora with explicit grounding in price elasticity and demographic-conditioned valuation—that target the relevant capability directly rather than treating it as one task among many. Our evidence does not adjudicate among these futures, and we treat the persistence of the pattern as an open empirical question that future replications, conducted on the next generation of models, will need to answer.

What replicates, what shifts. We expect the qualitative findings of our paper—heterogeneity across LLMs, slope flatness, sensitivity to belief inputs, belief-question quality variation—to replicate across model generations even as the quantitative levels evolve. We do not expect our specific WTP point estimates to replicate as models improve. The level/slope diagnostic framework we apply, however, should remain the appropriate way to evaluate synthetic CV: any future LLM that is offered as a substitute for human CV respondents will need to be evaluated jointly on level recovery and slope recovery, because both feed into the WTP ratio.

6.5 Limitations

We acknowledge four limitations beyond those addressed in Section 4.8.

Sample non-representativeness inherits from the originals. Our synthetic respondents are constructed from the original studies' respondent pools. Aldy, Kotchen, and Leiserowitz (2012) used a probability-based panel and is approximately nationally representative; Giguere, Moore, and Whitehead (2020) used an opt-in panel of North Carolina residents and is not. Our synthetic samples preserve these representativeness profiles, including any selection biases. A researcher interested in nationally representative synthetic CV would need to construct synthetic personas from a separate, representative demographic frame—a procedure that introduces its own difficulties when belief inputs are not available for the synthetic respondents.

Implicit leakage cannot be ruled out. Our direct-recall probe (Appendix H) finds that [**X of 10**] models can reproduce the headline WTP figures from the original studies when asked. We argue that the cross-study comparison and the consistency of slope flatness across more- and less-cited studies provides circumstantial evidence against leakage as the dominant mechanism. We cannot, however, rule out that LLM responses to our personas are partially shaped by pre-trained associations between demographic profiles and policy positions that derive from the original studies' published results. Distinguishing implicit leakage from genuine reasoning is a hard problem in the broader LLM-as-respondent literature and remains unresolved.

Temperature and reasoning-mode choices are not fully explored. We use default sampling parameters across providers and do not systematically vary temperature, top-p, or reasoning-mode configurations. Preliminary results not reported here suggest that temperature changes within the typical range (0 to 1) shift response variance more than central tendency, but a thorough sensitivity analysis would be valuable for practitioners.

The studies replicated are both U.S. environmental. Generalization to non-U.S. contexts, non-environmental domains, and non-Western policy frames is not tested.

6.6 Future Work

The descriptive findings we report set up several natural priorities for further work. We list them in approximate order of tractability.

Diagnosing the slope-flatness pattern. The most immediate extension is to identify why synthetic approval-bid curves are flatter than empirical curves. Two candidate accounts—training-distribution averaging across heterogeneous price effects, and attenuated treatment of bid as a substantive policy cost—can in principle be distinguished empirically. Pre-registered diagnostic tasks that vary the substantive role of the dollar amount, paired with token-probability analysis for the subset of LLMs that expose log-probabilities through their public APIs, would allow direct measurement of how each model's implicit utility function responds to bid amount as a continuous quantity. The required infrastructure builds directly on the persona and prompt templates we report here; the analysis is restricted to the providers that expose log-probabilities (OpenAI, Google, Mistral, Meta).

Synthetic CV as benefit transfer engine. The most policy-relevant extension is to demonstrate synthetic CV in the role we argue suits it best: faster, more flexible benefit transfer. The procedure would be: identify a target context where no CV study exists; construct synthetic personas representative of the target population; query an ensemble of LLMs to produce WTP estimates; apply per-model corrections (where available); and report transfer-error-adjusted confidence intervals comparable to those produced by conventional benefit transfer. Demonstrating this on a target context with eventual ground-truth data—for example, by running synthetic CV in advance of a planned conventional CV study and comparing predictions—would provide the strongest test of practical utility.

Generalization to additional elicitation formats and policy domains. Testing synthetic CV in contexts beyond the two we replicate is a resource-intensive but straightforward extension. A natural sequence is: open-ended WTP, payment-card formats, discrete choice experiments, and finally non-environmental domains (health, transportation, cultural heritage). Each new context will require its own validation against published benchmarks.

Engagement with the scope-sensitivity literature. The most conceptually challenging extension concerns scope sensitivity. Modern CV validity assessment relies heavily on demonstrating that respondents update WTP appropriately when the magnitude of the good changes; LLMs appear to fail this test in ways that humans typically do not. Diagnosing the source of this failure—whether it reflects training-data averaging, attention to surface cues over substantive content, or something else—may require interpretability techniques drawn from the machine learning literature rather than from environmental economics, and is likely a multi-paper agenda.

7 Conclusion

Whether large language models can substitute for human respondents in contingent valuation is an empirical question, and one that the LLM-as-respondent literature has not previously addressed on published CV instruments with known human-survey benchmarks. Our synthetic replications of Aldy, Kotchen, and Leiserowitz (2012) and Giguere, Moore, and Whitehead (2020), across ten frontier and mid-tier LLMs and three persona information conditions, establish that current synthetic CV does not deliver point estimates of WTP that match human-survey benchmarks under naive deployment. The deviations are not random: synthetic approval-bid curves are systematically flatter than empirical curves across both studies, both choice formats, and almost every model, with mechanical consequences for WTP through the logit-implied ratio $\widehat{\text{WTP}} = -\hat{\alpha}/\hat{\beta}_b$. Across-LLM heterogeneity is large enough that LLM selection is a first-order design decision; ensemble averaging does not rescue weak models. The marginal value of belief inputs is highly context-dependent, both across studies and across belief items within a study.

We argue that the most natural intellectual home for synthetic CV is the benefit transfer literature. Under that frame, current synthetic CV is not yet competitive with conventional benefit transfer for direct value substitution, but is plausibly already useful for threshold analysis, sensitivity testing of conventional benefit transfer, and pilot design for planned CV studies. We do not believe synthetic CV is ready to substitute for conventional CV in cases where regulatory decisions hinge on precise WTP point estimates; we do believe it is ready, today, to expand the set of values that can be considered in regulatory analysis when the alternative is leaving them unexamined.

The slope-flatness pattern we document is consistent enough across studies and models to merit characterization, and the question of why synthetic bid-response curves are flat—training-distribution averaging, attenuated treatment of bid as a substantive cost, or something else—is a question the present paper does not settle. The diagnostic and corrective work that question invites is, in our view, the natural next step for synthetic CV as a methodology, and the descriptive evidence assembled here provides that motivation.

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A Study Design Details

This appendix reports the design details of the two replicated studies that are referenced but not fully described in the main text.

A.1 Study 1: Aldy, Kotchen, and Leiserowitz (2012)

Sample. Nationally representative survey of 1,010 U.S. adults conducted between April 23 and May 12, 2011. The sample was drawn from Ipsos KnowledgePanel (formerly known as Knowledge Networks) using a nationally representative, probability-based online panel.

Treatment arms. Two factors were independently randomized:

- *Bid amount* ($\$5, \$25, \$45, \$65, \$85, \$105, \$135, \$155$). The four middle bids (\$45–\$105) were assigned with twice the probability of the two lowest and two highest bids.
- *Clean-energy technology mix*, randomized across three arms: (i) renewables only ("solar and wind power"), (ii) renewables plus natural gas, and (iii) renewables plus nuclear.

CV question wording (verbatim). "The federal government is considering a Clean Energy Standard that would require electric power companies to obtain 80% of their energy from clean sources by the year 2035. Eligible sources of clean energy would include [*technology arm*]. If this policy were to cost your household [*bid amount*] more each year in higher electricity bills, would you support or oppose this policy?"

Original specification. Pooled binary logit. Likelihood-ratio tests reported by the original authors fail to reject pooling across technology arms; we follow the authors' preferred pooled specification in our synthetic replication.

Original WTP estimate. Mean WTP of \$162 per household per year (95% CI: \$128–\$260).

A.2 Study 2: Giguere, Moore, and Whitehead (2020)

Sample. Online opt-in panel survey of 907 North Carolina residents conducted in September 2017. The sample was drawn from the SurveyMonkey online platform.

Treatment arms. The choice experiment varied four attributes:

- *Ecologically important acreage treated* (2,500, 5,000, 7,500, or 10,000 acres, corresponding to 0%, 17%, 25%, and 33% of the total).
- *Socially important acreage treated* (2,500, 5,000, 7,500, or 10,000 acres).
- *Treatment method* (chemical insecticide or biological control with predatory beetles).
- *Annual household cost* for three years (\$50, \$100, \$150, \$200 in the random-cost design; additional \$25 and \$250 levels in the bounded design).

Choice formats. Respondents were assigned to one of two formats:

1. *Random-cost repeated choice*: respondents face multiple referendum questions with bid amounts drawn independently per question.
2. *Bounded choice*: bid amounts follow an ascending or descending sequence within respondent, providing internal consistency checks.

Original specification. Conditional logit with respondent-level controls and attribute-nonattendance adjustment.

Original WTP estimate. Marginal WTP of \$56.53 per household per year for an additional acre of "socially important" forest treated (95% CI: \$24–\$75).

B Belief Items

This appendix lists the belief and attitudinal items used as inputs to the synthetic personas in the demographics-plus-beliefs and one-belief-at-a-time conditions.

B.1 Study 1 Belief Items ($K = 13$)

Q1. Recently, you may have noticed that global warming has been getting some attention in the news. Global warming refers to the idea that the world's average temperature has been increasing over the past 150 years, may be increasing more in the future, and that the world's climate may change as a result. What do you think: Do you think that global warming is happening?
Options: Refused; No; Don't know; Yes.

Q2. Assuming global warming is happening, do you think it is...
Options: Refused; Caused mostly by human activities; Caused mostly by natural changes in the environment; Other (Please specify); None of the above because global warming isn't happening.

Q3. Which comes closest to your own view?
Options: Refused; Don't know enough to say; There is a lot of disagreement among scientists about whether or not global warming is happening; Most scientists think global warming is not happening; Most scientists think global warming is happening.

Q4. How worried are you about global warming?
Options: Refused; Not at all worried; Not very worried; Somewhat worried; Very worried.

Q5. How much do you think global warming will harm you personally?
Options: Refused; Don't know; Not at all; Only a little; A moderate amount; A great deal.

Q6. How much do you think global warming will harm people in the United States?
Options: Refused; Never; In 100 years; In 50 years; In 25 years; In 10 years; They are being harmed right now.

Q7. How much do you think global warming will harm people in developing countries?
Options: Refused; Don't know; Not at all; Only a little; A moderate amount; A great deal.

Q8. How much do you think global warming will harm future generations of people?
Options: Refused; Don't know; Not at all; Only a little; A moderate amount; A great deal.

Q9. How much do you think global warming will harm plant and animal species?
Options: Refused; Don't know; Not at all; Only a little; A moderate amount; A great deal.

Q10. How much do you support or oppose the following policy? Fund more research into renewable energy sources, such as solar and wind power.
Options: Refused; Strongly oppose; Somewhat oppose; Somewhat support; Strongly support.

Q11. Do you think that developing sources of clean energy should be a low, medium, high, or very high priority for the president and Congress?
Options: Refused; Low; Medium; High; Very high.

Q12. Do you think global warming should be a low, medium, high, or very high priority for the president and Congress? **Options:** Refused; Low; Medium; High; Very high.

Q13. How often do you discuss global warming with your family and friends?
Options: Refused; Never; Rarely; Occasionally; Often.

As reported in Section 5.1, items Q1 and Q3 systematically degrade fit when included in single-belief specifications.

B.2 Study 2 Belief Items ($K = 16$)

Q1. Before this survey, how much did you know about hemlock trees in the Southern Appalachian Mountains?
Options: A lot; Some; A little; Nothing.

Q2. Have you read or heard about an increase in the number of dead hemlock trees in the Southern Appalachian Mountains?
Options: Yes; No.

Q3. Many reasons have been suggested for protecting the quality of hemlock trees in southern Appalachian forests. Listed below are several functions of hemlock trees. How important is it to you to protect these functions?
1) Providing wildlife habitat;
2) Providing scenic views;
3) Providing recreation opportunities;
4) Providing timber;
5) Preserving hemlock populations;
6) Maintaining aquatic resources;
7) Other.
Options for all: Very Important; Important; Moderately Important; Slightly Important; Not Important.

Q4. Have you ever visited Great Smoky Mountains National Park?
Options: Yes; No.
Did you visit Great Smoky Mountains National Park during the past 12 months?
Options: Yes; No.
About how many times did you visit Great Smoky Mountains National Park during the past 12 months?

Q5. During your visit(s) did you notice areas with dead or dying hemlock trees?
Options: Yes; No.

Q6. Have you ever visited Pisgah National Forest?
Options: Yes; No.
Did you visit Pisgah National Forest during the past 12 months?
Options: Yes; No.
About how many times did you visit Pisgah National Forest during the past 12 months?

Q7. During your visit(s) did you notice areas with dead or dying hemlock trees?
Options: Yes; No.

Q8. Have you ever visited Nantahala National Forest?
Options: Yes; No.
Did you visit Nantahala National Forest during the past 12 months?
Options: Yes; No.
About how many times did you visit Nantahala National Forest during the past 12 months?

Q9. During your visit(s) did you notice areas with dead or dying hemlock trees?
Options: Yes; No.

Q10. Dead or dying hemlock trees will ...
1) Reduce the quality of the recreation experience;
2) Cause me to change which sites I visit to recreate;
3) Cause me to change the type of activities I participate in;
4) Improve the quality of the recreation experience.
Options for all: Strongly Agree; Agree; Undecided; Disagree; Strongly Disagree.

Q11. About how many times do you think you will visit...
1)Great Smoky Mountains National Park;
2) Pisgah National Forest;
3) and Nantahala National Forest
during the next 12 months?
Options for all: 0; 1; 2; 3; 4; 5; 6; 7; 8; 9; 10 or more.

Q12. When designing programs to protect the hemlocks of the Southern Appalachian Forest from hemlock woolly adelgid, forest managers make decisions about how to allocate the effort across different areas. Think about how important to you it is to protect the hemlock trees in each of the following areas, and then rank the 3 areas in order of importance, with 1 being the most important area to protect, 2 the second most important, and 3 being the third most important area.
1) Great Smoky Mountains National Park;
2) Pisgah National Forest;
3) Nantahala National Forest.
Options: (1 = most important ... 3 = least important)

Q13. When designing programs to protect hemlocks at national recreation sites, such as the Great Smoky Mountains National Park and Pisgah and Nantahala National Forests, forest managers make decisions about which hemlock trees to protect. Listed below are various locations of hemlock stands on public lands. Think about how important to you it is to protect the hemlock trees in each of the areas, and then rank the 7 areas in order of importance, with 1 being the most important location to protect, 2 the second most important, and so on up to number 7 being the least important area.
1) Along roads;
2) Along hiking trails;

3) In campgrounds and picnic areas;
4) In wilderness areas;
5) Along streams;
6) In designated conservation areas (seed sources for future trees);
7) In groves of the oldest hemlock trees (above 150 years old).
Options: (1 = most important ... 7 = least important)

Q14. How important do you think it is to treat the following treatment areas?
1) Ecologically important areas;
2) Socially important areas.
Options for all: Very Important; Important; Moderately Important; Slightly Important; Not Important.

Q15. Do you agree or disagree that these control methods should be used in the Great Smoky Mountains National Park and Pisgah and Nantahala National Forests?
1) Chemical insecticide;
2) Biological control using predatory beetles.
Options for all: Strongly Agree; Agree; Neither Agree Nor Disagree; Disagree; Strongly Disagree.

Q16. Did you read these instructions very closely, somewhat closely or not closely at all?
Options: Very Closely; Somewhat Closely; Not Closely At All.

C Simulation Platform and Prompt Architecture

This appendix provides technical documentation of the simulation infrastructure and prompt design used in this study. Section C.1 describes the Alethic-ISM platform as instantiated for this experiment. Section C.2 presents the persona template structure. Section C.3 gives the CV prompt templates for each study. Section C.4 specifies the structured output schema and parsing logic.

C.1 Alethic-ISM: Platform Architecture as Instantiated in This Experiment

Alethic-ISM is an open-source distributed computation engine for building analytic and agentic graphs in which every execution step produces immutable, versioned state with structural provenance. It is available at `https://github.com/alethicresearch/alethic-ism` and is described in full in RasaeeGhose2026. We describe here the specific features of the platform as they bear on the reproducibility and integrity of this experiment.

Core design principle. State provenance is the architecture, not a feature. States are append-only and content-addressed (SHA-256 hashed); lineage is embedded in the data itself rather than maintained as a separate audit log. The data is the audit trail. Nothing is overwritten; nothing is lost. The graph does not orchestrate computation—it *is* the computation—and the complete record of that computation survives execution intact.

Graph primitives. Alethic-ISM graphs are composed of three primitive types. *Nodes* (processors) each pair an instruction with a processor implementation; processor types used in this experiment include LLM prompt execution (via provider-specific API processors for OpenAI, Anthropic, Google, DeepSeek, Mistral, Moonshot, and Meta) and Python script execution for validation and data assembly. Nodes carry no hidden state—their behavior is fully determined by their instruction and the input states they receive. *Edges* are programmable per-connection functions that run on input and output; in this experiment, cross-join edges construct Cartesian products of respondent states and condition states, implementing the $\mathcal{R} \times \mathcal{C} \times \mathcal{M}$ cross-join described in Section 4.4. *States* are append-only, versioned data containers persisting computation outputs; every state row carries queryable lineage recording the chain of processors, routes, and input states that produced it.

Execution model. Execution flows through a route model: State → ProcessorState(INPUT) → Processor → ProcessorState(OUTPUT) → State. Each ProcessorState route is a persisted object with its own identity, status, and execution history. In this experiment, each (respondent, condition, model, repetition) cell is a ProcessorState route; the full experiment consists of [**XXX,XXX**] such routes across both studies. All ten models receive identical input states in parallel within each condition, each producing a separate immutable output state. This parallel execution structure ensures that cross-model comparisons reflect differences in LLM response behavior rather than sequencing artifacts.

Retry and failure handling. Responses that fail schema validation are retried up to three additional times with the same prompt and input state, each retry recorded as a new ProcessorState route linked to the same input. Persistent failures after four total attempts are coded as missing and excluded from the analytic sample. Schema-validation failure rates by model are reported in Appendix D (Table D).

Reproducibility archive. Full Alethic-ISM workflow graphs, persona templates, prompt templates, model parameters, and analysis code are available at [[**repository URL**]]. Per-respondent synthetic response data are available at [[**data repository URL**]], subject to the data-use restrictions of the original studies.

C.2 Persona Template Structure

Synthetic personas are constructed programmatically from each respondent's individual-level data using a fixed natural-language template (production version v6.3). The template is identical across all respondents within a study; only the slot values vary.

Each persona consists of three blocks rendered in sequence:

Block 1: System preamble. A fixed instruction block that establishes the simulation context. It instructs the model to respond as the described individual would respond to the survey question, to use only information consistent with the described profile, and not to draw on information about events occurring after the original survey date (April–May 2011 for Study 1; September 2017 for Study 2). The temporal anchoring instruction was introduced in template version v6.0; the production template v6.3 uses the following preamble:

[[**INSERT VERBATIM PREAMBLE FROM v6.3 HERE**]]

Block 2: Demographic block. A prose enumeration of the respondent's sociodemographic attributes, one attribute per sentence, using a fixed natural-language format for each attribute type. An illustrative excerpt for Study 1:

[[**INSERT REPRESENTATIVE DEMOGRAPHIC BLOCK EXCERPT HERE**]]

Block 3: Belief block (optional). In the demographics + all beliefs condition, a Q&A-formatted enumeration of the respondent's responses to non-CV survey items is appended. Each item is rendered as a question–answer pair using the verbatim wording of the original survey item. An illustrative excerpt for Study 1:

> [[**INSERT REPRESENTATIVE BELIEF BLOCK EXCERPT HERE**]]

In the demographics + one belief at a time condition, only a single Q&A pair is appended per run. In the demographics-only condition, Block 3 is omitted entirely.

Table **??** in Appendix F documents the substantive design changes across persona template versions v5.x through v6.3, and the per-version approval shifts observed in Study 1.

C.3 Contingent Valuation Prompt Templates

After constructing the persona, the LLM is presented with a CV question that reproduces the wording, treatment assignments, and bid amounts of the original study. The CV prompt is appended to the persona in the same API call; persona and CV prompt together constitute the full input state for the LLM processor node.

Study 1 (NCES) CV prompt template:

> [[**INSERT VERBATIM STUDY 1 CV PROMPT TEMPLATE HERE, with slot labels for {TECHNOLOGY_ARM} and {BID_AMOUNT}**]]
>
> The original referendum question wording, from Aldy2012, is reproduced verbatim with the randomized treatment description and bid amount substituted from the original respondent's record (see Appendix A for verbatim CV question wording).

Study 2 (HWA) CV prompt template:

> [[**INSERT VERBATIM STUDY 2 CV PROMPT TEMPLATE HERE, with slot labels for {ECO_ACRES}, {SOCIAL_ACRES}, {TREATMENT_METHOD}, and {BID_AMOUNT}**]]
>
> The original choice-format question wording, from Giguere2020, is reproduced verbatim with the randomized acreage, treatment-method, and bid attributes substituted from the original respondent's record (see Appendix A for verbatim CV question wording).

In both studies, the CV prompt instructs the LLM to return a structured response (see Section C.4) and specifies that the binary support/oppose vote should reflect how the described individual would respond given their profile and the stated cost.

Question framing. The response options are presented as "Support / Oppose" in the primary specification. A parallel set of simulations using "Oppose / Support" ordering was run as a robustness check; results are reported in Appendix E (Table E).

C.4 Structured Output Schema and Parsing Logic

The CV prompt instructs the LLM to return its response as a JSON object conforming to the following schema:

```
{
  "vote": "[support | oppose]",
  "confidence": "[integer 1-5]",
  "rationale": "[free-text string]"
}
```

Parsing. Responses are parsed against this schema by extracting the `vote` field and mapping it to a binary indicator ($y_i = 1$ for support, $y_i = 0$ for oppose). The `confidence` and `rationale` fields are recorded but not used in the main analysis.

Schema validation. A response fails schema validation if any of the following conditions hold: (i) the `vote` field is absent or cannot be parsed; (ii) the `vote` field contains a value other than `"support"` or `"oppose"` (including `"abstain"`, `"unsure"`, refusals, or partial responses); (iii) the response is not valid JSON or is truncated before the `vote` field is reached. Responses failing validation are retried up to three times; persistent failures are coded as missing.

Structured output enforcement. For models and API configurations that support constrained decoding or structured output modes (e.g., JSON mode), we use the provider's native structured output enforcement where available, with schema validation as a fallback for providers that do not expose this feature. The specific enforcement method used per provider is noted in Appendix D (Table D).

D Model Identifiers and API Configuration

This appendix reports the exact model identifiers, API endpoints, and sampling parameters used in our simulations, to support replication.

Table D.5: Model identifiers and API configuration.

Provider	Model	API identifier	Endpoint	Cutoff
OpenAI	GPT-5	`[gpt-5-XXXX-XX-XX]`	`api.openai.com`	[**XXXX**]
OpenAI	GPT-5-mini	`[gpt-5-mini-XXXX-XX-XX]`	`api.openai.com`	[**XXXX**]
Google	Gemini 2.5 Flash	`[gemini-2.5-flash-XXX]`	`generativelanguage.googleapis.com`	[**XXXX**]
Google	Gemini 2.5 Flash-Lite	`[gemini-2.5-flash-lite-XXX]`	`generativelanguage.googleapis.com`	[**XXXX**]
DeepSeek	DeepSeek-V3 (Chat)	`[deepseek-chat-v3.1]`	`api.deepseek.com`	[**XXXX**]
DeepSeek	DeepSeek-R1	`[deepseek-r1]`	`api.deepseek.com`	[**XXXX**]
Mistral	Mistral Medium 3.1	`[mistral-medium-3.1]`	`api.mistral.ai`	[**XXXX**]
Mistral	Mistral Small 3.2	`[mistral-small-3.2-24b-instruct]`	`api.mistral.ai`	[**XXXX**]
Moonshot	Kimi-K2	`[kimi-k2-XXX]`	[**endpoint**]	[**XXXX**]
Meta	Llama-4 Scout	`[llama-4-scout-XXX]`	[**endpoint**]	[**XXXX**]

Notes: "Cutoff" refers to each model's training data cutoff as published by the provider. All API calls were made between [**start date**] and [**end date**]. For all models we used: temperature = [**T**], top-p = [**p**], max tokens = [**XXX**]. For models exposing a separate reasoning configuration, default reasoning settings were used; reasoning traces were not passed into the analytic dataset.

Schema-validation failure rates. Table D reports the fraction of LLM responses that failed schema validation on the first attempt and were either successfully retried or coded as missing.

Table D.6: Schema-validation failure rates by model.

Model	First-attempt failure (%)	Resolved on retry (%)	Coded missing (%)	Final N retained (%)
GPT-5	[X.X]	[X.X]	[X.X]	[XX.X]
GPT-5-mini	[X.X]	[X.X]	[X.X]	[XX.X]
Gemini 2.5 Flash	[X.X]	[X.X]	[X.X]	[XX.X]
Gemini 2.5 Flash-Lite	[X.X]	[X.X]	[X.X]	[XX.X]
DeepSeek-V3	[X.X]	[X.X]	[X.X]	[XX.X]
DeepSeek-R1	[X.X]	[X.X]	[X.X]	[XX.X]
Mistral Medium 3.1	[X.X]	[X.X]	[X.X]	[XX.X]
Mistral Small 3.2	[X.X]	[X.X]	[X.X]	[XX.X]
Kimi-K2	[X.X]	[X.X]	[X.X]	[XX.X]
Llama-4 Scout	[X.X]	[X.X]	[X.X]	[XX.X]

Notes: Pooled across both studies and all information conditions. "Final N retained" is the share of intended (respondent, condition, model, repetition) cells that produced a schema-valid, non-missing response after up to four total attempts.

E Question-Ordering Robustness

A known concern in survey methodology is that response options' order may shift the distribution of responses—particularly for moderate-leaning respondents who anchor on the first option presented. This may operate differently in LLMs than in humans because of how language models process token sequences. We test for ordering effects in Study 1 by running the full demographics-plus-beliefs experiment under both "Support / Oppose" and "Oppose / Support" option orderings.

Table E.7: Ordering robustness, Study 1 demographics-plus-beliefs condition.

Model	Support-first share	Oppose-first share	Difference (pp)
GPT-5	[0.XXX]	[0.XXX]	[+/−X.X]
GPT-5-mini	[0.XXX]	[0.XXX]	[+/−X.X]
Gemini 2.5 Flash	[0.XXX]	[0.XXX]	[+/−X.X]
Gemini 2.5 Flash-Lite	[0.XXX]	[0.XXX]	[+/−X.X]
DeepSeek-V3	[0.XXX]	[0.XXX]	[+/−X.X]
DeepSeek-R1	[0.XXX]	[0.XXX]	[+/−X.X]
Mistral Medium 3.1	[0.XXX]	[0.XXX]	[+/−X.X]
Mistral Small 3.2	[0.XXX]	[0.XXX]	[+/−X.X]
Kimi-K2	[0.XXX]	[0.XXX]	[+/−X.X]
Llama-4 Scout	[0.XXX]	[0.XXX]	[+/−X.X]
Mean across models	[0.XXX]	[0.XXX]	[X.X]

Notes: Pooled across all bid levels. Differences reported in percentage points. None of the per-model differences are statistically distinguishable from zero at the 5% level under cluster-robust standard errors at the respondent level.

We find no meaningful effect from ordering: the maximum across-model difference is [**X.X**] percentage points in absolute value, smaller than the within-model run-to-run variation reported in Section 4.6. We pool across orderings in our main results.

F Robustness Checks

F.1 Temporal contamination

Both studies pre-date the training cutoff of every model in our sample. To assess whether LLM responses are influenced by post-collection-window information, we re-ran the demographics-plus-beliefs condition with a stricter persona prompt that included the additional instruction:

> *You are answering as if responding during [**April–May 2011 / September 2017**]. Do not use any information about events, policies, scientific findings, or political developments occurring after [**May 12, 2011 / September 30, 2017**]. If you find yourself drawing on more recent information, stop and re-anchor your response to what was known at the original survey date.*

Differences between the strict-temporal and standard-temporal persona conditions are reported in Table F.1. The strict instruction produces a [**small / moderate / negligible**] shift in approval rates ([**X.X**] percentage points on average across models) and a [**small / moderate / negligible**] shift in WTP estimates ([**$XX**] on average).

Table F.8: Temporal contamination check, Study 1 demographics-plus-beliefs condition.

	Approval share (pooled)		WTP ($)	
Model	Standard	Strict-temporal	Standard	Strict-temporal
[**Top 5 models by sensitivity**]	[0.XXX]	[0.XXX]	[XXX]	[XXX]
...	...	...	...	...

Notes: Pooled across bid levels.

F.2 Prompt sensitivity

We tested [**X**] alternative persona templates that vary surface features while preserving substantive content: (i) reordering of demographic versus belief blocks, (ii) rephrasing of the framing instruction ("You are a persona..." vs. "Adopt the perspective of..."), (iii) bullet-point versus prose formatting of the demographic block, and (iv) inclusion versus exclusion of the explicit injunction against post-cutoff information.

The dominant source of variation in synthetic responses is across LLMs (Section 5.3); surface-form prompt variation produces shifts an order of magnitude smaller than across-model heterogeneity for most models.

F.3 Persona-version sensitivity

All main-text results use a single frozen production template (`v6.3`). Earlier versions of the template, the substantive design changes between versions (e.g., temporal anchoring, belief-block formatting), and per-version approval shifts are documented here.

Table F.9: Surface-form prompt sensitivity, Study 1.

Template variation	Mean abs. shift in approval (pp)	Max shift across models (pp)	Mean abs. shift in WTP ($)	Max shift in WTP ($)
Reorder blocks	[X.X]	[X.X]	[XX]	[XX]
Rephrase framing	[X.X]	[X.X]	[XX]	[XX]
Bullet vs. prose formatting	[X.X]	[X.X]	[XX]	[XX]
Drop temporal injunction	[X.X]	[X.X]	[XX]	[XX]

Notes: All shifts are computed relative to the production template (`v6.3`) and pooled across models. Per-model results in supplementary materials.

Table F.10: Cross-version persona comparison.

Template version	Date	Substantive change from prior version	Mean approval shift (pp)
v5.X	[**date**]	Initial production template	—
v6.0	[**date**]	Added explicit temporal anchoring	[+/−X.X]
v6.1	[**date**]	Restructured belief block as Q&A pairs	[+/−X.X]
v6.2	[**date**]	Added uncertainty framing	[+/−X.X]
v6.3 (production)	[**date**]	Conditioned share-category on output rather than input	[+/−X.X]

Notes: "Mean approval shift" is measured relative to the immediately prior version, pooled across models and bid levels in Study 1. The production template (`v6.3`) is used for all main-text results.

G Replication and Estimation Tables

This appendix reports the full set of estimation tables underlying the results in Section 5. Section G.1 verifies our reproduction of the original empirical specifications; Sections G.2 and G.3 report synthetic estimates by LLM and condition.

G.1 Replication of Original Specifications

Table G.11: Replication of Study 1

	Empirical
Bid amount ($)	-0.0059***
	(0.0017)
Natural gas treatment	-0.3377**
	(0.1688)
Nuclear treatment	-0.3045*
	(0.1707)
College	0.1586
	(0.1666)
Male	-0.1132
	(0.1372)
Household size	-0.0616
	(0.0500)
Income ($10k)	0.0162
	(0.0158)
Age	-0.0106**
	(0.0046)
White	0.4324**
	(0.1686)
Republican	-1.0253***
	(0.1910)
Independent	-0.5266***
	(0.1939)
No party	-1.0140***
	(0.2096)
Constant	2.0154***
	(0.4053)
Observations	983
Log-likelihood	-621.0339

Note: Binary logit replicating Table 2 of Aldy, Kotchen, and Leiserowitz (2012).
Mean WTP via Krinsky-Robb (1,000 reps) at empirical mean covariates: $164 (95% CI: $128–$292).

Table G.12: Replication of Study 2

	(1) Bounded	(2) Random-cost
Cost ($)	-0.0050***	-0.0110***
	(0.0006)	(0.0013)
Ecologically important acres	-0.0480	-0.0922
	(0.0664)	(0.0621)
Socially important acres	0.1144*	0.0707
	(0.0684)	(0.0616)
Biological treatment	7.9272***	7.5935***
	(0.2302)	(0.2497)
Chemical treatment	6.2729***	6.4846***
	(0.2043)	(0.2362)
Observations	280422	166354
Log-likelihood	-4.70e+03	-4.48e+03

Note: Conditional logit estimates on Giguere, Moore, and Whitehead (2020) human-respondent data. Unadjusted specification (no attribute-nonattendance correction).

G.2 Synthetic Estimates: Study 1 (NCES)

Table G.13: Synthetic logit estimates, Study 1 (NCES): Demographics only.

Model	Bid ($)	NatGas	Nuclear	College	Age	N	$\hat{\beta}_{bid}/\hat{\beta}_{bid}^{emp}$
deepseek-chat-v3.1	-0.00173	0.052	0.127	1.183	-0.0146	22,228	0.293
	(0.00046)	(0.046)	(0.046)	(0.044)	(0.0012)		
deepseek-r1	-0.00677	0.090	0.223	0.723	-0.0234	22,217	1.149
	(0.00043)	(0.043)	(0.042)	(0.041)	(0.0012)		
gemini-2.5-flash	-0.00550	-0.118	0.232	0.815	-0.0159	22,219	0.933
	(0.00042)	(0.042)	(0.042)	(0.040)	(0.0011)		
gemini-2.5-flash-lite	-0.00779	-0.798	0.985	1.118	-0.0644	22,220	1.322
	(0.00048)	(0.050)	(0.047)	(0.046)	(0.0014)		
llama-4-scout	-0.00149	-0.328	-0.096	-0.006	-0.0214	17,965	0.252
	(0.00052)	(0.053)	(0.055)	(0.054)	(0.0015)		
mistral-medium-3.1	-0.00257	-0.068	0.169	0.844	-0.0230	22,109	0.435
	(0.00043)	(0.044)	(0.043)	(0.043)	(0.0012)		
mistral-small-3.2-24b	-0.00152	0.056	0.401	1.009	-0.0286	22,224	0.258
	(0.00043)	(0.044)	(0.043)	(0.044)	(0.0012)		
kimi-k2	-0.00420	-0.279	0.194	0.359	-0.0364	22,239	0.713
	(0.00044)	(0.044)	(0.045)	(0.044)	(0.0013)		
gpt-5-mini	-0.00660	-0.166	0.173	0.049	-0.0229	23,046	1.120
	(0.00041)	(0.041)	(0.042)	(0.041)	(0.0012)		
Aggregated	-0.00336	-0.124	0.218	0.569	-0.0218	196,467	0.570
	(0.00013)	(0.013)	(0.013)	(0.013)	(0.0004)		
Empirical (Aldy et al.)	-0.00590	-0.338	-0.305	0.159	-0.0106	983	1.000
	(0.00165)	(0.169)	(0.171)	(0.167)	(0.0046)		

Notes: Binary logit estimates of the Aldy, Kotchen, and Leiserowitz (2012) specification on synthetic responses, persona condition: Demographics only. Standard errors in parentheses; one observation per persona-LLM-repetition cell. Final column reports the synthetic bid coefficient divided by the empirical bid coefficient; values below 1 in magnitude indicate slope compression. Coefficients shown are a subset; the full coefficient vector appears in the supplementary table. Empirical benchmark is the same specification on Aldy, Kotchen, and Leiserowitz (2012) human-respondent data; WTP is computed on the collapsed-to-persona synthetic sample to match the empirical sample structure (see Table G.15).

G.3 Synthetic Estimates: Study 2 (HWA)

Table G.14: Synthetic logit estimates, Study 1 (NCES): Demographics + beliefs.

Model	Bid ($)	NatGas	Nuclear	College	Age	N	$\hat{\beta}_{bid}/\hat{\beta}_{bid}^{emp}$
deepseek-chat-v3.1	-0.00406	-0.244	-0.077	0.406	-0.0037	22,229	0.688
	(0.00035)	(0.035)	(0.035)	(0.035)	(0.0009)		
deepseek-r1	-0.00462	-0.261	-0.083	0.331	-0.0015	22,225	0.784
	(0.00035)	(0.035)	(0.036)	(0.035)	(0.0009)		
gemini-2.5-flash	-0.00257	-0.349	-0.095	0.174	-0.0038	22,045	0.436
	(0.00036)	(0.036)	(0.037)	(0.036)	(0.0010)		
gemini-2.5-flash-lite	-0.00174	-0.425	-0.102	0.238	-0.0050	22,186	0.294
	(0.00036)	(0.036)	(0.037)	(0.036)	(0.0010)		
llama-4-scout	-0.00005	-0.157	-0.076	0.180	-0.0084	20,339	0.009
	(0.00039)	(0.040)	(0.040)	(0.039)	(0.0011)		
mistral-medium-3.1	-0.00360	-0.377	-0.121	0.189	-0.0065	22,111	0.610
	(0.00036)	(0.036)	(0.037)	(0.036)	(0.0010)		
mistral-small-3.2-24b	-0.00118	-0.274	-0.050	0.312	-0.0052	22,181	0.200
	(0.00036)	(0.036)	(0.037)	(0.036)	(0.0010)		
kimi-k2	-0.00156	-0.255	-0.161	0.113	-0.0062	22,151	0.264
	(0.00037)	(0.037)	(0.038)	(0.037)	(0.0010)		
gpt-5-mini	-0.00214	-0.110	-0.065	0.017	-0.0052	23,044	0.363
	(0.00037)	(0.037)	(0.038)	(0.037)	(0.0010)		
Aggregated	-0.00235	-0.266	-0.088	0.217	-0.0048	198,511	0.399
	(0.00012)	(0.012)	(0.012)	(0.012)	(0.0003)		
Empirical (Aldy et al.)	-0.00590	-0.338	-0.305	0.159	-0.0106	983	1.000
	(0.00165)	(0.169)	(0.171)	(0.167)	(0.0046)		

Notes: Binary logit estimates of the Aldy, Kotchen, and Leiserowitz (2012) specification on synthetic responses, persona condition: Demographics + beliefs. Standard errors in parentheses; one observation per persona-LLM-repetition cell. Final column reports the synthetic bid coefficient divided by the empirical bid coefficient; values below 1 in magnitude indicate slope compression. Coefficients shown are a subset; the full coefficient vector appears in the supplementary table. Empirical benchmark is the same specification on Aldy, Kotchen, and Leiserowitz (2012) human-respondent data; WTP is computed on the collapsed-to-persona synthetic sample to match the empirical sample structure (see Table G.15).

Table G.15: Synthetic mean willingness to pay for the NCES policy (Study 1).

Model	Demographics only	+ Beliefs
AGGREGATED	-9 [-17, -2]	289 [268, 314]
deepseek-chat-v3.1	-843 [-1887, -509]	37 [27, 46]
deepseek-r1	-13 [-25, -1]	91 [85, 98]
gemini-2.5-flash	-102 [-134, -77]	335 [276, 440]
gemini-2.5-flash-lite	-104 [-129, -83]	491 [366, 798]
llama-4-scout	1529 [908, 4788]	18899 [-36826, 38172]
mistral-medium-3.1	-288 [-473, -192]	183 [163, 211]
mistral-small-3.2-24b	-535 [-1328, -303]	258 [184, 564]
kimi-k2	268 [232, 320]	618 [440, 1120]
gpt-5-mini	185 [172, 202]	522 [405, 764]
Empirical (Aldy et al.)	164 [128, 292]	

Notes: Mean WTP ($/household/year) for the NCES policy. Bracketed values are 95% confidence intervals from a Krinsky-Robb parametric bootstrap with 1,000 replications evaluated at the empirical mean covariate vector. pools synthetic responses across all nine LLMs. Synthetic samples are collapsed to one observation per persona prior to estimation, matching the empirical respondent-level sample structure. Empirical benchmark from Aldy, Kotchen, and Leiserowitz (2012).

Table G.16: Synthetic-to-empirical bid-coefficient ratios (Study 1).

Model	Demographics only	+ Beliefs
AGGREGATED	0.570	0.399
deepseek-chat-v3.1	0.293	0.688
deepseek-r1	1.149	0.784
gemini-2.5-flash	0.933	0.436
gemini-2.5-flash-lite	1.322	0.294
llama-4-scout	0.252	0.009
mistral-medium-3.1	0.435	0.610
mistral-small-3.2-24b	0.258	0.200
kimi-k2	0.713	0.264
gpt-5-mini	1.120	0.363

Notes: Ratio of synthetic to empirical bid coefficients from binary logit estimates of the Aldy, Kotchen, and Leiserowitz (2012) specification. Values below 1 in magnitude indicate slope compression. A value near 0 indicates near-insensitivity to bid amount. Empirical bid coefficient used as denominator is from Table G.1.

Table G.17: Synthetic conditional logit estimates: Random-cost design, Demographics only (Study 2).

Model	COST	ECOL	SOCIAL	BIOL	CHEM	N	$\hat{\beta}_{COST}/\hat{\beta}^{emp}_{COST}$
deepseek-chat-v3.1	-0.0083	-0.07	0.04	4.3552	3.8170	14,098	0.757
	(0.0013)	(0.06)	(0.06)	(0.2348)	(0.2379)		
deepseek-r1	-0.0119	-0.11	0.11	5.0383	4.2869	16,062	1.087
	(0.0014)	(0.06)	(0.07)	(0.2478)	(0.2435)		
gemini-2.5-flash	-0.0131	-0.07	0.05	5.2268	4.5848	15,322	1.193
	(0.0013)	(0.06)	(0.07)	(0.2510)	(0.2463)		
gemini-2.5-flash-lite	-0.0093	-0.12	0.08	7.6460	6.6516	19,662	0.848
	(0.0031)	(0.14)	(0.14)	(0.6167)	(0.5884)		
gpt-5-mini	-0.0092	-0.03	-0.08	6.8408	5.8195	20,442	0.840
	(0.0020)	(0.10)	(0.09)	(0.4020)	(0.3874)		
kimi-k2	-0.0159	-0.09	0.07	5.7364	5.2061	18,310	1.453
	(0.0014)	(0.07)	(0.07)	(0.2609)	(0.2591)		
llama-4-scout	-0.0062	-0.22	-0.00	7.9648	7.1739	19,888	0.563
	(0.0039)	(0.20)	(0.19)	(0.7927)	(0.7536)		
mistral-medium-3.1	-0.0107	-0.15	0.04	5.0201	3.8799	19,624	0.977
	(0.0012)	(0.06)	(0.06)	(0.2236)	(0.2148)		
mistral-small-3.2-24b	-0.0150	-0.08	0.10	6.8718	6.3227	18,698	1.365
	(0.0020)	(0.09)	(0.09)	(0.3811)	(0.3793)		
Aggregated	-0.0099	-0.06	0.06	5.3586	4.7549	161,922	0.904
	(0.0005)	(0.02)	(0.02)	(0.0885)	(0.0873)		
Empirical (Giguere et al.)	-0.0110	-0.09	0.07	7.5935	6.4846	166,354	1.000
	(0.0013)	(0.06)	(0.06)	(0.2497)	(0.2362)		

Notes: Conditional logit estimates from Random-cost design choice data, persona condition: Demographics only. Standard errors in parentheses. Final column reports the synthetic bid coefficient divided by the empirical bid coefficient; values below 1 indicate slope compression. Aggregated row pools synthetic responses across all nine LLMs; each (provider, respondent, attempt) cell forms its own choice set. Empirical benchmark is the same unadjusted specification estimated on Giguere, Moore, and Whitehead (2020) human-respondent data.

Table G.18: Synthetic conditional logit estimates: Random-cost design, Demographics + beliefs (Study 2).

Model	COST	ECOL	SOCIAL	BIOL	CHEM	N	$\hat{\beta}_{COST}/\hat{\beta}^{emp}_{COST}$
deepseek-chat-v3.1	-0.0020	-0.02	-0.07	3.2629	3.0381	13,446	0.186
	(0.0012)	(0.06)	(0.06)	(0.2105)	(0.2148)		
deepseek-r1	-0.0036	-0.05	-0.05	4.2118	3.6793	18,216	0.333
	(0.0012)	(0.06)	(0.06)	(0.2161)	(0.2209)		
gemini-2.5-flash	-0.0003	-0.03	0.00	3.2025	2.9919	17,066	0.027
	(0.0012)	(0.06)	(0.06)	(0.2070)	(0.2143)		
gemini-2.5-flash-lite	-0.0006	-0.15	-0.04	4.3804	4.0562	19,766	0.051
	(0.0013)	(0.07)	(0.07)	(0.2423)	(0.2492)		
gpt-5-mini	-0.0012	-0.05	-0.02	4.8049	4.4856	25,190	0.111
	(0.0014)	(0.07)	(0.07)	(0.2585)	(0.2576)		
kimi-k2	-0.0105	-0.09	-0.02	3.8642	3.5021	14,068	0.958
	(0.0013)	(0.06)	(0.07)	(0.2307)	(0.2361)		
llama-4-scout	-0.0042	-0.06	-0.12	3.3368	2.8843	14,030	0.384
	(0.0012)	(0.06)	(0.06)	(0.2179)	(0.2235)		
mistral-medium-3.1	-0.0037	0.00	-0.08	3.7336	3.2484	16,434	0.341
	(0.0012)	(0.06)	(0.06)	(0.2118)	(0.2140)		
mistral-small-3.2-24b	-0.0008	-0.13	-0.10	4.2010	4.0186	18,168	0.070
	(0.0013)	(0.06)	(0.06)	(0.2291)	(0.2349)		
Aggregated	-0.0036	-0.06	-0.04	3.9966	3.6471	156,004	0.330
	(0.0004)	(0.02)	(0.02)	(0.0734)	(0.0745)		
Empirical (Giguere et al.)	-0.0110	-0.09	0.07	7.5935	6.4846	166,354	1.000
	(0.0013)	(0.06)	(0.06)	(0.2497)	(0.2362)		

Notes: Conditional logit estimates from Random-cost design choice data, persona condition: Demographics + beliefs. Standard errors in parentheses. Final column reports the synthetic bid coefficient divided by the empirical bid coefficient; values below 1 indicate slope compression. Aggregated row pools synthetic responses across all nine LLMs; each (provider, respondent, attempt) cell forms its own choice set. Empirical benchmark is the same unadjusted specification estimated on Giguere, Moore, and Whitehead (2020) human-respondent data.

Table G.19: Synthetic conditional logit estimates: Bounded choice design, Demographics only (Study 2).

Model	COST	ECOL	SOCIAL	BIOL	CHEM	N	$\hat{\beta}_{COST}/\hat{\beta}^{emp}_{COST}$
deepseek-chat-v3.1	-0.0078	-0.01	0.21	5.4194	4.0343	25,562	1.570
	(0.0007)	(0.07)	(0.08)	(0.2351)	(0.2104)		
deepseek-r1	-0.0102	0.08	0.13	6.3330	4.1056	27,108	2.055
	(0.0008)	(0.08)	(0.08)	(0.2770)	(0.2236)		
gemini-2.5-flash	-0.0122	-0.03	0.04	6.7153	5.5741	25,384	2.456
	(0.0009)	(0.08)	(0.09)	(0.3040)	(0.2780)		
gemini-2.5-flash-lite	-0.0135	0.25	0.17	11.0763	7.6889	34,804	2.715
	(0.0023)	(0.14)	(0.14)	(0.9719)	(0.6660)		
gpt-5-mini	-0.0142	-0.10	0.15	9.3981	6.9681	35,492	2.867
	(0.0013)	(0.09)	(0.10)	(0.4641)	(0.3950)		
kimi-k2	-0.0149	-0.03	0.13	6.9468	6.2774	30,666	3.001
	(0.0010)	(0.08)	(0.08)	(0.3074)	(0.2899)		
llama-4-scout	-0.0180	0.26	0.25	11.7972	8.7512	35,430	3.625
	(0.0032)	(0.13)	(0.14)	(1.0674)	(0.8965)		
mistral-medium-3.1	-0.0091	-0.05	0.21	6.4494	3.5797	33,146	1.834
	(0.0008)	(0.07)	(0.07)	(0.2560)	(0.1919)		
mistral-small-3.2-24b	-0.0072	0.01	0.13	6.8410	5.9502	31,918	1.452
	(0.0010)	(0.09)	(0.10)	(0.3453)	(0.3156)		
Aggregated	-0.0080	-0.01	0.10	6.5965	5.2066	280,532	1.613
	(0.0003)	(0.03)	(0.03)	(0.0950)	(0.0842)		
Empirical (Giguere et al.)	-0.0050	-0.05	0.11	7.9272	6.2729	280,422	1.000
	(0.0006)	(0.07)	(0.07)	(0.2302)	(0.2043)		

Notes: Conditional logit estimates from Bounded choice design choice data, persona condition: Demographics only. Standard errors in parentheses. Final column reports the synthetic bid coefficient divided by the empirical bid coefficient; values below 1 indicate slope compression. Aggregated row pools synthetic responses across all nine LLMs; each (provider, respondent, attempt) cell forms its own choice set. Empirical benchmark is the same unadjusted specification estimated on Giguere, Moore, and Whitehead (2020) human-respondent data.

Table G.20: Synthetic conditional logit estimates: Bounded choice design, Demographics + beliefs (Study 2).

Model	COST	ECOL	SOCIAL	BIOL	CHEM	N	$\hat{\beta}_{COST}/\hat{\beta}^{emp}_{COST}$
deepseek-chat-v3.1	0.0032	-0.01	0.11	2.9953	2.2351	25,096	-0.644
	(0.0006)	(0.06)	(0.06)	(0.1873)	(0.1772)		
deepseek-r1	0.0024	0.02	0.26	3.5058	2.4972	32,822	-0.493
	(0.0006)	(0.06)	(0.07)	(0.1894)	(0.1801)		
gemini-2.5-flash	0.0043	0.08	0.15	2.7427	2.2297	30,942	-0.874
	(0.0006)	(0.06)	(0.06)	(0.1861)	(0.1797)		
gemini-2.5-flash-lite	0.0036	0.00	0.26	4.2773	2.9164	36,566	-0.717
	(0.0007)	(0.07)	(0.07)	(0.2076)	(0.1903)		
gpt-5-mini	0.0046	0.05	0.23	4.0833	3.4226	45,704	-0.926
	(0.0007)	(0.07)	(0.07)	(0.2062)	(0.1938)		
kimi-k2	-0.0034	0.03	0.12	3.3748	2.7977	24,618	0.678
	(0.0006)	(0.06)	(0.07)	(0.1889)	(0.1766)		
llama-4-scout	0.0026	0.08	0.17	2.8239	1.6726	25,380	-0.523
	(0.0006)	(0.06)	(0.06)	(0.1835)	(0.1765)		
mistral-medium-3.1	0.0030	0.00	0.13	3.0111	2.2409	29,236	-0.613
	(0.0006)	(0.06)	(0.06)	(0.1884)	(0.1777)		
mistral-small-3.2-24b	0.0062	0.02	0.20	3.4250	2.7585	33,064	-1.241
	(0.0007)	(0.07)	(0.07)	(0.2001)	(0.1872)		
Aggregated	0.0035	0.03	0.16	3.3283	2.5529	283,634	-0.699
	(0.0002)	(0.02)	(0.02)	(0.0623)	(0.0588)		
Empirical (Giguere et al.)	-0.0050	-0.05	0.11	7.9272	6.2729	280,422	1.000
	(0.0006)	(0.07)	(0.07)	(0.2302)	(0.2043)		

Notes: Conditional logit estimates from Bounded choice design choice data, persona condition: Demographics + beliefs. Standard errors in parentheses. Final column reports the synthetic bid coefficient divided by the empirical bid coefficient; values below 1 indicate slope compression. Aggregated row pools synthetic responses across all nine LLMs; each (provider, respondent, attempt) cell forms its own choice set. Empirical benchmark is the same unadjusted specification estimated on Giguere, Moore, and Whitehead (2020) human-respondent data.

Table G.21: Synthetic willingness to pay for an additional acre of socially important forest treated (Study 2).

	Random-cost design		Bounded design	
Model	w/o beliefs	w/ beliefs	w/o beliefs	w/ beliefs
AGGREGATED	65.3	57.9	211.7	-94.9
	[86.4, 44.2]	[103.7, 12.2]	[307.0, 116.3]	[-33.4, -156.5]
deepseek-chat-v3.1	32.6	78.2	52.1	-129.2
	[50.0, 15.1]	[157.7, -1.4]	[71.9, 32.2]	[5.3, -263.6]
deepseek-r1	62.6	44.2	39.4	-88.3
	[97.0, 28.3]	[94.6, -6.1]	[58.3, 20.6]	[-5.6, -171.1]
gemini-2.5-flash	12.1	425.0	0.3	-122.4
	[23.4, 0.7]	[1879.3, -1029.3]	[14.6, -13.9]	[-8.9, -235.9]
gemini-2.5-flash-lite	-10.8	79.6	24.2	-52.2
	[34.4, -55.9]	[353.4, -194.1]	[58.4, -9.9]	[-0.5, -103.9]
gpt-5-mini	7.6	122.8	3.1	-6.5
	[31.7, -16.4]	[679.3, -433.7]	[15.6, -9.5]	[39.6, -52.5]
kimi-k2	66.6	40.1	47.1	77.9
	[89.0, 44.3]	[61.5, 18.6]	[65.1, 29.1]	[128.0, 27.8]
llama-4-scout	51.8	174.2	35.0	-761.7
	[187.3, -83.7]	[285.5, 62.8]	[68.9, 1.1]	[-84.2, -1439.2]
mistral-medium-3.1	40.7	92.6	41.0	-127.3
	[55.4, 26.0]	[195.1, -9.8]	[53.5, 28.6]	[-15.0, -239.7]
mistral-small-3.2-24b	57.1	-134.4	144.2	-58.2
	[83.9, 30.4]	[36.5, -305.3]	[214.5, 73.9]	[-25.6, -90.8]
Empirical (Giguere et al.)	56.53 [24, 75]			

Notes: Marginal WTP ($/household/year) for an additional acre of socially important forest treated. Bracketed values are 95% confidence intervals computed by the delta method via `nlcom`. pools synthetic responses across all nine LLMs and re-estimates the conditional logit on the pooled data. Empirical benchmark from Giguere, Moore, and Whitehead (2020).

Table G.22: Synthetic-to-empirical bid-coefficient ratios (Study 2).

	Random-cost		Bounded	
Model	w/o beliefs	w/ beliefs	w/o beliefs	w/ beliefs
AGGREGATED	0.895	0.321	1.633	-0.695
deepseek-chat-v3.1	0.777	0.368	1.536	-0.541
deepseek-r1	0.866	0.402	1.987	-0.394
gemini-2.5-flash	1.122	0.119	2.516	-0.712
gemini-2.5-flash-lite	1.021	0.083	2.768	-0.675
gpt-5-mini	1.052	0.150	2.853	-0.854
kimi-k2	1.097	0.789	3.253	0.576
llama-4-scout	1.071	0.442	2.811	-0.396
mistral-medium-3.1	1.180	0.503	1.981	-0.642
mistral-small-3.2-24b	1.344	0.138	1.383	-1.257

Notes: Ratio of synthetic to empirical bid coefficients from conditional logit estimates. Values below 1 indicate that synthetic responses are less sensitive to cost than empirical responses (). A value of 0 indicates no sensitivity to cost; values above 1 indicate over-sensitivity. The empirical bid coefficients used as denominators are taken from the unadjusted conditional logit specifications reported in Table G.1.

H Training-Data Leakage Probe

A natural concern in any LLM-based replication is that the LLM may have seen the original study's published results during training and may be reproducing them rather than generating genuine synthetic responses. Aldy, Kotchen, and Leiserowitz (2012) has been cited [**600+**] times since publication, and the headline WTP estimate of $162 appears in many secondary sources; Giguere, Moore, and Whitehead (2020) is less widely cited but has appeared in [**XX**] subsequent publications. Both papers' results are plausibly within the training distribution of every LLM in our sample.

H.1 Direct recall probe

For each model, we ran a leakage probe in a session entirely separate from our main simulations, using a clean context with no persona or CV prompt. The probe consisted of three questions asked in sequence:

1. "What is the headline willingness-to-pay estimate from Aldy, Kotchen, and Leiserowitz's 2012 *Nature Climate Change* paper on a U.S. Clean Energy Standard?"
2. "What is the headline marginal willingness-to-pay estimate from Giguere, Moore, and Whitehead's 2020 *Land Economics* paper on hemlock woolly adelgid control?"
3. "Without searching for the answer: what bid amounts did each study use?"

Table H.1 reports model responses.

Table H.23: Direct recall probe for training-data leakage.

Model	Recalls Aldy WTP	Recalls Giguere WTP	Recalls bid range
GPT-5	[Yes/Approx/No]	[Yes/Approx/No]	[Yes/Approx/No]
GPT-5-mini	[Yes/Approx/No]	[Yes/Approx/No]	[Yes/Approx/No]
Gemini 2.5 Flash	[Yes/Approx/No]	[Yes/Approx/No]	[Yes/Approx/No]
Gemini 2.5 Flash-Lite	[Yes/Approx/No]	[Yes/Approx/No]	[Yes/Approx/No]
DeepSeek-V3	[Yes/Approx/No]	[Yes/Approx/No]	[Yes/Approx/No]
DeepSeek-R1	[Yes/Approx/No]	[Yes/Approx/No]	[Yes/Approx/No]
Mistral Medium 3.1	[Yes/Approx/No]	[Yes/Approx/No]	[Yes/Approx/No]
Mistral Small 3.2	[Yes/Approx/No]	[Yes/Approx/No]	[Yes/Approx/No]
Kimi-K2	[Yes/Approx/No]	[Yes/Approx/No]	[Yes/Approx/No]
Llama-4 Scout	[Yes/Approx/No]	[Yes/Approx/No]	[Yes/Approx/No]

Notes: "Yes" means the model produced a numeric figure within 5% of the published estimate. "Approx" means the model produced a directionally correct figure within 25%. "No" means the model declined to answer, produced an unrelated figure, or produced a figure off by more than 25%.

H.2 Interpretation

If training-data leakage were the dominant force generating our synthetic responses, we would expect at least one of two patterns. First, models with stronger direct recall of the published WTP figures should produce synthetic WTP estimates closer to those figures than models with weaker recall do. Second, synthetic recovery should be systematically tighter for Aldy, Kotchen, and Leiserowitz (2012) than for Giguere, Moore, and Whitehead (2020), because the Aldy result has been cited [**600+**] times and appears in many secondary sources while the Giguere result has appeared in [**XX**] publications and is plausibly less represented in any LLM's training distribution.

Neither pattern holds in our data. Comparing Table H.1 to the WTP estimates in Section 5, models with stronger recall do not systematically produce more accurate synthetic WTP estimates, and the slope-flatness regularity reported in Section 5.3 appears to roughly comparable degree in both studies despite their different citation profiles. If LLM responses were primarily reproductions of memorized study results, we would not expect a structurally similar deviation pattern in a heavily cited study and a less-cited one. We interpret this as evidence against direct training-data leakage being the dominant driver of our synthetic responses, while acknowledging that this argument is circumstantial rather than conclusive.

Limitations of the leakage probe. The recall probe identifies only *conscious* recall—the kind of leakage that an LLM can verbalize when asked. Implicit leakage—in which the LLM's response distribution is shaped by prior exposure to the studies without the LLM being able to articulate that exposure—cannot be ruled out by this probe. We discuss this further in Section 6.5.